

Reforming China's public pension system: Fiscal sustainability and the challenge of formality-based inequality^{☆,☆☆}

Han Gao^{id}*, Bei Lu

UNSW School of Economics, Australia

ARTICLE INFO

Dataset link: [Replication Package for Reforming China's Public Pension System: Fiscal Sustainability and the Challenge of Formality-Based Inequality \(Original data\)](#)

JEL classification:

D15

H55

O17

Keywords:

China

Pension reform

Informality

ABSTRACT

This paper examines the design of China's public pension system, addressing two major challenges: fiscal unsustainability caused by population aging and significant pension disparities between formal and informal workers. Using a dynamic quantitative model with long-run macro and demographic trends, we assess the effects of the ongoing reforms and offer several options to deepen the reform: indexing pension benefits solely to the CPI, raising the resident pensions, and replacing pension financing from payroll tax with consumption tax. Our findings indicate that the delayed retirement policy announced in September 2024 could substantially mitigate fiscal deficits in the coming decades. Among the options evaluated, raising resident pension alleviates the formality-based pension inequality but increases fiscal burden; altering benefit indexation or the tax base for pension financing enhances both fiscal sustainability and welfare outcomes; the effects of consumption tax are particularly pronounced in the long run.

1. Introduction

China's pension system plays a vital role in social and economic policy and has been fundamentally transformed in recent decades. Although significant progress has been made in expanding coverage, it remains a system where different groups of workers (formal/informal, migrant/local, urban/rural, men/women) have significantly different levels of financial protection. And, like other countries that are rapidly ageing, China's pension system is subject to growing fiscal pressures with implications not only for the well-being of older people, but also for overall fiscal balances.

In this paper, we build a dynamic quantitative model to understand how reforming the public pension system can help address the sustainability and equity of the system. We first examine the impacts of the 2024 reform and examine its impact on public finance and household behavior responses including labor supply and savings. Building on

the 2024 policy as the new baseline, we propose and evaluate several options aimed at further addressing policy concerns.

In the past 30 years, China has successfully expanded its pension coverage to nearly the entire population through two major schemes: the Enterprise Basic Pension Scheme (EBPS) and the Rural and Urban Residents Pension Scheme (RURPS).¹ Adjustments to these schemes have followed trends in ageing and the need for economic development in the country. By 2023, more than 521 million were enrolled in EBPS (including 142 million retirees), representing approximately half of the labor force. The remainder were enrolled in RURPS, which had 545 million members, including 173 million retirees.

The EBPS system is challenged by the increasing fiscal budget pressure driven by legacy costs and an ageing demographic structure. Although the pension system includes an individual account component, it primarily operates on a pay-as-you-go (PAYG) basis. The individual accounts function as notional defined contribution accounts,

[☆] This article is part of a Special issue entitled: 'Population aging' published in Economic Modelling.

^{☆☆} We thank the editor Sushanta Mallick and anonymous referees for their constructive suggestions. We also thank John Piggot and Philip O'Keefe for their useful insights when preparing for the DRC/ADB research program on high quality growth in China, Charles Yuji Horioka and Aiko Kikkawa for helpful discussions, and the support of the computational facility of the National Computational Infrastructure (NCI) through the UNSW HPC Scheme (DOI:10.26190/PMN5-7J50).

* Corresponding author.

E-mail addresses: han.gao6@unsw.edu.au (H. Gao), lubei@unsw.edu.au (B. Lu).

¹ We use EBPS to refer to the newly merged one that covers employees in for-profit enterprises and non-profit government institutions. Prior to 2015, the EBPS only covered workers employed in for-profit enterprises, and the PEP (Public Employee Pension) covered civil servants and employees in non-profit government institutions. In 2015, the PEP was merged with the old EBPS, making the new EBPS the uniform program for all employees in urban sectors.

offering a credit rate of return that has exceeded bank fixed interest rates over the past decade. Social pooling contributions are based on a formula that incorporates the average wage factor and years of contribution.

Although EBPS coverage has made notable progress in the past decade, increasing from 25% in 2010 to more than half of the workforce, further expansion faces several hurdles. The labor force has been shrinking since 2018, while the number of retirees is increasing significantly. Slower economic growth may put a pressure on firms, which face high pension costs, to employ fewer high-cost workers. In addition, advances in technology, particularly artificial intelligence, may replace certain jobs, further complicating efforts to expand the formal employment sector. In addition to overall budgetary challenges (Sin, 2005; Song et al., 2015), regional disparities in economic conditions pose a significant fiscal challenge for local governments (Li and Lin, 2019).

To maintain fiscal sustainability of EBPS, the government announced plans to gradually increase the pension eligibility age for formal workers in September 2024. The retirement age will increase from 50 to 55 for blue-collar female workers, from 55 to 58 for white-collar female workers, and from 60 to 63 for male workers. Early or late retirement options, allowing flexibility of up to three years, will also be introduced. Although much progress has been made, significant inequality remains between formal and informal workers. Currently, the benefits provided by RURPS are only about 6% of those provided by EBPS. Therefore, further reforms are expected to address the financial sustainability of EBPS and the disparity between EBPS and RURPS.

To understand how pension reform options can affect the aggregate economy and household behavior, we conduct long-run simulation experiments using a quantitative model. Building on a heterogeneous agent model where households make endogenous choices of labor supply and consumption subject to idiosyncratic shocks and incomplete markets, the model is extended to allow features that match the salient characteristics of China, such as intergenerational transfers between generations, labor income processes that differ by formal and informal and formality-based public pension policy. The model is further extended to include long-run macro- and demographic trends, including population aging, productivity slowdown, and formalization of the population.

Our policy experiments are conducted in three steps. First, we evaluate the impacts of the delaying retirement policy announced in 2024 on government budget, household savings, and labor supply. Second, we treat the new policy as the baseline and propose several options of further reforms as possible solutions to maintain long-term fiscal sustainability and alleviate the disparities between formal and informal workers. In the first option, we consider a change in pension indexation from a combination of full CPI indexation and partial wage indexation to CPI-indexation only. Option 2 involves an increase in the resident pension, i.e., a higher pension replacement ratio for the informal workers. In option 3, we consider an alternative pension financing scheme where the current wage contribution rate of 16% for social pooling and 8% for individual accounts is replaced with individual accounts funded by consumption tax at the rate of 25%. In all experiments, we examine the impacts on aggregates, such as government fiscal balances, the impacts on welfare by formality, and the impacts on household behaviors such as labor supply, consumption, and savings. Finally, to shed further light on the role of model assumptions in shaping the results, we conduct robustness checks of policy impact on government budget and welfare along several dimensions: fertility rate projection, intergenerational transfers, long-run wage growth, internal returns of pension accounts, and preference parameters.

We show that the delaying retirement policy recently announced in September 2024 can significantly alleviate the issue of fiscal deficits in the next few decades. While the reform only modestly mitigates overall demographic trends by increasing employment by 30 million (6% of employment), it substantially affects the fiscal budget. To be

more specific, the pension deficit still grows, but the annual deficit will fall from 9.5% to 5.8% of GDP by 2075. Discounted using the market rate, the cumulative discounted present value (DPV) of the government balance will be lowered from 390% to 210% of GDP.

Among the three further reform options, we find that further removing wage indexation of pension benefits improves the welfare the most under our baseline specification. When pension benefits are indexed to CPI only, the government runs smaller deficits and thus can reimburse the saved revenue to households either in the form of lump-sum transfers or a reduced pension tax rate, under which agents make better intertemporal decisions when they are young. Raising pension benefits for informal workers redistributes from formal to informal workers, which inevitably worsens formal workers' welfare. Yet, it still improves overall welfare for cohorts born before 2020 since informal workers have a higher marginal utility of consumption. The consumption tax financing scheme increases the disposable income for formal workers and equitably includes informal workers into the system. Despite its distortion to intertemporal decisions, it still improves the welfare of both groups in the long run. As these options address different aspects of the system, policy makers can consider a combination of them.

We perform thorough robustness checks along several dimensions of the model to verify the validity of the results. Under the low fertility rate projection, the government runs larger deficits, and the fiscal costs saved by further reform options are also expanded, leading to larger welfare effects. The presence of intergenerational transfers weakens the insurance value of public pension programs, in particular, for the informal workers, who rely more on the transfers from their children. As a consequence, removing intergenerational transfers leads to an overestimation of welfare gains of reforms, especially for informal workers. A lower long-run wage growth will decrease the contribution of the latter cohorts, exacerbating the pension fund deficits under a PAYG system and having an impact of welfare through its impact on the size of transfers and contribution taxes to facilitate the transition. A lower discount factor or a higher risk aversion imply a stronger incentive for agents to receive transfers or contribution tax cuts as young as possible, altering the welfare effects of several reform options as well.

Related literature and contribution. We make three contributions to the literature. First, we build a framework with incomplete market and endogenous savings and labor supply choices by heterogeneous households that suits the empirical background of China. We enrich the stylized model (İmrohoroglu and Kitao, 2012; Kitao, 2014) in two dimensions: the large disparity between formal and informal workers and the long-run challenges faced by the public pension system due to the aging of the population and the slowdown of productivity. Previous literature on Chinese pension reform has highlighted the roles of elements such as endogenous fertility (Coeurdacier et al., 2014), rural-urban mobility (Bairaliya and Miller, 2021), and health (Deng et al., 2023), but they only focus on steady-state comparisons and, therefore, are not suitable for evaluating policy in a long-run transitional dynamic environment. An exception is (Song et al., 2015), which highlights the role of productivity growth in shaping the welfare effects between cohorts. Relative to Song et al. (2015), we incorporate richer elements such as formality-based inequality, intergenerational transfers, and long-run demographic and macro trends.

Second, we study the pension reform that was recently announced in September 2024 demonstrating the strong policy relevance of our study. We provide timely estimates on impacts of the reform and evaluate several routes for deepening the reform. Similar works have been carried out in countries such as the United States (Conesa and Krueger, 1999; İmrohoroglu and Kitao, 2012), Spain (Díaz-Giménez and Díaz-Saavedra (2009), and Japan (Kitao, 2014, 2015). Compared to them, we make policy proposals that are deeply rooted in Chinese characteristics—a representative emerging economy featuring large inequality and unsustainability of the public pension system in transition.

For example, the option to change pension benefits indexation is proposed as a continuation of the reform previously implemented in 2005, and the option to increase resident pension is aimed at addressing the formality-based inequality that is unique in developing countries including China.

Lastly, we conduct thorough sensitivity checks along the dimensions of various aspects of the model. Our results highlight the quantitative importance of incorporating elements with Chinese features such as fertility and intergenerational transfers (Banerjee et al., 2023; Choukhmane et al., 2023) and long-run productivity growth (Song et al., 2015). We also provide estimates of policy impact under alternative choices of pension account returns and preference parameters, establishing the quantitative validity of our results.

The structure of the paper is as follows. Section 2 provides a brief review of the transformation of the Chinese public pension system and raises the two key challenges facing it. Section 3 then presents an overlapping generations (OLG) model with long-run dynamics of demographic and macro trends. In Section 4, we use the model to project the future trajectory of the pension system in a pre-2024 reform scenario and compare it to the 2024 reform in terms of impacts on public finances and household decisions. Section 5 discusses the policy implications of the analysis and suggests several options to deepen the reform of the Chinese pension system. Section 6 checks the robustness under alternative assumptions of model elements such as fertility, intergenerational transfers, and long-run trends. Section 7 concludes.

2. The Chinese public pension system: Transformation and challenges

The Chinese public pension system has undergone several rounds of reform in recent decades that have fundamentally transformed the landscape of pensions in the country.²

From 1951, pensions, or “labor insurance”—were provided on an unfunded basis by SOEs and collectively owned enterprises in urban areas, with a parallel Public Employee Pension scheme. Although finances were pooled at the local level during the 1980s, enterprises continued to be the source of funding, and the system was PAYG.³ However, the market transition and SOE restructuring increasingly made the enterprise-based model unsuitable for China’s needs.

In this context, the 1997 reform was a major milestone for China’s pension system, moving to a social insurance model for urban workers with a PAYG pooled defined benefit (DB) pillar and a defined contribution (DC) individual account (EBPS). Civil servants continued to have a parallel scheme. However, these schemes covered only workers in the formal urban sector and the public sector, leaving a major coverage gap for workers in the informal urban and rural sectors. Migrant workers were also not included in the early years of the urban workers’ scheme until the 2008 Labor Contract Law and the 2011 Social Insurance Law introduced an in-principle mandate for employers to pay their contributions (Meng, 2017). In addition to the core schemes for urban workers, firms were encouraged to provide enterprise annuities (supplementary occupational schemes), and in recent years the authorities have promoted voluntary third-pillar private pensions. The other major development of the system was the introduction from 2009 of a unique hybrid scheme, first for rural workers and then urban informal workers, and now merged for the two groups, the RURPS. The scheme design combines elements of a matching defined contribution scheme (MDC) with an individual account, and a basic or social pension, and to our knowledge is unique in the world.

Coverage expansion has been substantial, both in the core EBPS and in the newer RURPS. As a result, by 2023, over 750 million workers —

over 80 percent of the working age population — were contributing to one of the country’s pension schemes. While significant progress has been made, two major challenges remain for the Chinese public pension system: fiscal unsustainability due to population ageing and large pension inequality between formal and informal workers.

2.1. Challenge 1: Population ageing

The first apparent challenge is the increasing dependency ratio due to population ageing. In Fig. 1(a), we plot the demographic process projected by the United Nations under the medium fertility scenario.

As Fig. 1(a) clearly shows, more than 15 million people will retire each year in the next 50 years. In particular, a massive increase in the number of pension fund recipients is projected to take place between 2020 and 2030 as the baby boomers born in the 1960s start to retire. The next two spikes are projected to occur in the 2050’s and 2070’s. In Fig. 1(b), we plot the ratio of age 60 and 65 years of age above to the total population. As Fig. 1(b) shows, the share of the population over 60 years of age was only around 10% in the early 2000s. In fact, the high growth in the past two decades has long been attributed to a low dependency ratio—the so-called “population dividend”. Eventually, the share of seniors has increased rapidly. By the end of the century, around half of the population will be over 60 years of age.

2.2. Challenge 2: Rural-urban disparity

Another prominent feature of the Chinese economy is the wide gap between the rural and urban regions in terms of many dimensions, including labor market returns and opportunities for participation in social programs.

The key indicators for EBPS and RURPS as of 2023 are outlined in Table 1.⁴ The figures highlight the massive size of the Chinese pension system (easily the largest in the world) and the impressive level of coverage achieved, but also the major remaining gap in financial protection between workers in the urban formal sector and those in the rural and urban informal sectors.

By 2023, the EBPS program had 521 million participants and the RURPS program had 545 million participants. Although the government has made efforts to promote universal coverage for all citizens, the gap in pension between urban and rural populations remains significant.

According to the 2010 census, right before the RURPS was established, only 10% of the rural elderly relied on public pension programs as their main source of income, which is in sharp contrast to the fact that 70% of the urban elderly received public pension. After the launch of RURPS, the public pension coverage for the rural elderly increased rapidly to 60% by 2018. However, the pension benefit level of RURPS still falls short compared to that of EBPS. As shown by Table 1, in the year 2023, the RP recipients, presumably mainly rural, received an amount of pension income that is only 6% of that of the EBPS recipients (2671/44,912). Compared to the generous benefit of the EBPS with a replacement ratio as high as 70% of wage income, the RURPS so far can only be considered as an old-age assistance to meet basic consumption needs.

For EBPS members, the generosity of pension benefits has declined over time. While average nominal benefits increased sharply during the first two decades of the century, the average replacement rate declined from a high starting point, falling rather sharply between 2000 and around 2011, before stabilizing at around 45 percent of average wages during the subsequent decade (Fig. 2(a)). Although the Ministry

² See O’Keefe et al. (2025) for a detailed summary of the policy background on the evolution and current situation of the Chinese pension system.

³ See Fang and Feng (2018) for more details about the reforms.

⁴ We categorize participants in the pension system into formal and informal groups regardless of rural or urban Hukou registration status. The formal sector includes those enrolled in EBPS, while the informal sector consists of RURPS members.

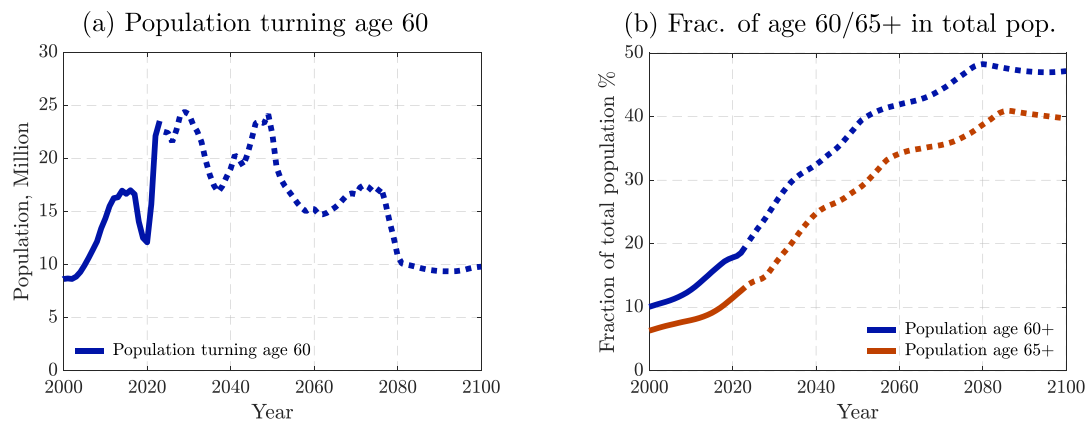


Fig. 1. Old age population over time.

Note: Data after 2023 are projected by [World Population Prospects 2023](#) under the medium fertility scenario published by the United Nations.

Table 1

Disposable incomes, pension, and consumption by formality in financial year 2023.

Data source: [2023 Statistical Bulletin on the Development of Human Resources and Social Security](#).

		Formal/EBPS	Informal/RURPS
Population, million		521.21	545.22
Contributors	Population, million	379.25	372.54
	Total contribution, billion CNY	70,506	6185
	Annual ave. contribution, CNY	18,590	1660
Recipients	Population, million	141.96	172.68
	Total benefits, billion CNY	63,757	4613
	Annual ave. benefits, CNY	44,912	2671
Incomes/consumption	Annual disposable incomes, CNY	51,821	21,691
	Annual wage incomes, CNY	31,321	9163
	Annual business & farm incomes, CNY	5903	7431
	Annual consumption, CNY	32,994	18,175

Notes: (i) Monetary variables are denominated in 2023 CNY. (ii) The average contribution is calculated per contributor, and the average benefit is calculated per benefit recipient. Variables in the income and consumption block, taken directly from the national bureau of statistics (NBS) of China, is denominated as per capita which includes all working and non-working population.

of Human Resources and Social Security(MOHRSS) had announced a target replacement rate of 59 percent of local average wage in 2005 for someone with 35 years of contributions ([Fang and Feng, 2018](#)), the average replacement rate by the early 2020s was still above the ILO minimum guidance of 40 percent.

One driver of the steady decline in the average replacement rate has been the way EBPS pension benefits have been adjusted over time. China has no strict indexation formula, but both prices and wages are taken into account in pension adjustments.⁵ [Fig. 2\(b\)](#) shows the trajectory of adjustments for 2005–2020, with adjustments in nearly all years markedly below average wage increases but well above CPI changes and until 2017 usually above the mean of wages and prices.

2.3. The 2024 reform

To maintain fiscal sustainability, the 11th Session of the Standing Committee of the 14th National People's Congress announcement was released in September 2024. Starting in 2025, retirement age will gradually be raised from 60 to 63 for men, from 50 to 55 for blue-collar women, and from 55 to 58 for white-collar women. [Fig. 3\(a\)](#) plots the

⁵ The 2005 reform changed the benefits formula from a multiplication of the target replacement ratio by the local average wage at retirement to a weighted average of the local average wage and the retiree's life-cycle average monthly wage. The average monthly wage is similar to the average indexed monthly earnings (AIME) in the US pension system. See [Salditt et al. \(2007\)](#) and [He et al. \(2019\)](#) for more details of the 2005 reform.

change in retirement age under the 2024 reform. Furthermore, starting in 2030, the minimum vesting period will be raised from 15 to 20 years by 2040. [Fig. 3\(b\)](#) provides the details.

3. Model

We develop an overlapping generation model with long-run dynamics, incorporated with demographic and macro labour and capital prices. Our framework closely follows earlier literature in this field, such as [Conesa and Krueger \(1999\)](#), [İmrohoroglu and Kitao \(2009, 2012\)](#) and [Kitao \(2014\)](#) and extends their framework by introducing the formal and informal sectors and sector-contingent public pension system.

The model unit of analysis is the average of a representative household member, which takes the average of a man and a woman within the same household.⁶ Agents are heterogeneous with a permanent formality status, stochastic income shocks and endogenous wealth.⁷

⁶ Modeling individual as the decision unit apparently brings advantage in describing individuals with richer heterogeneity, such as gender, but it may still fail to account for the transfers across generations and across household members. See [Gao \(2025\)](#) for a model with costless transfers across generations and men and women household members.

⁷ We make this modeling choice because formality is tightly linked to local 'hukou' status that is fixed for an individual over life. Most of the change in 'hukou' status is achieved by acquiring more advanced education.

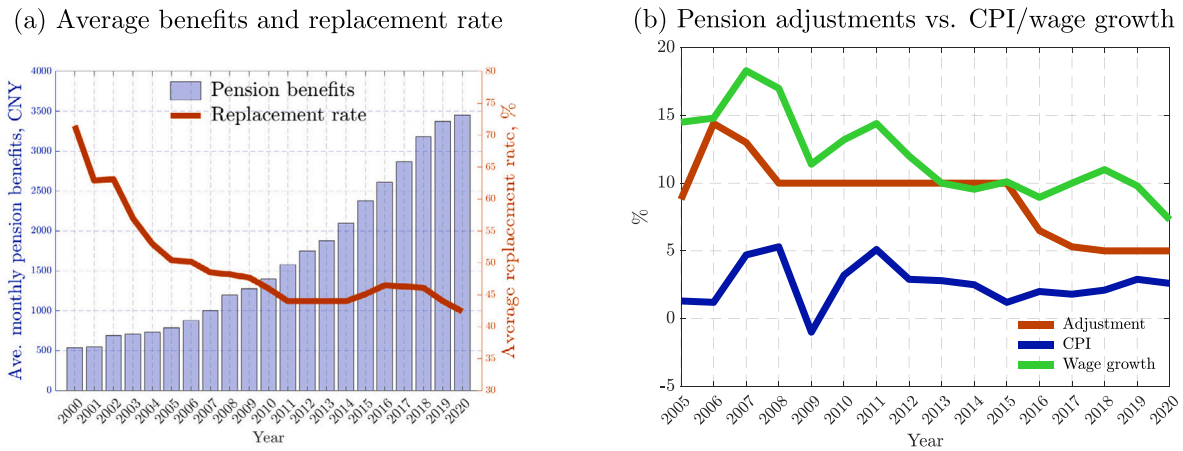


Fig. 2. EBPS pension benefits and indexation over time. Note: Data sources: (Wang and Feng, 2021), based on NBS, MOHRSS and State Council data.

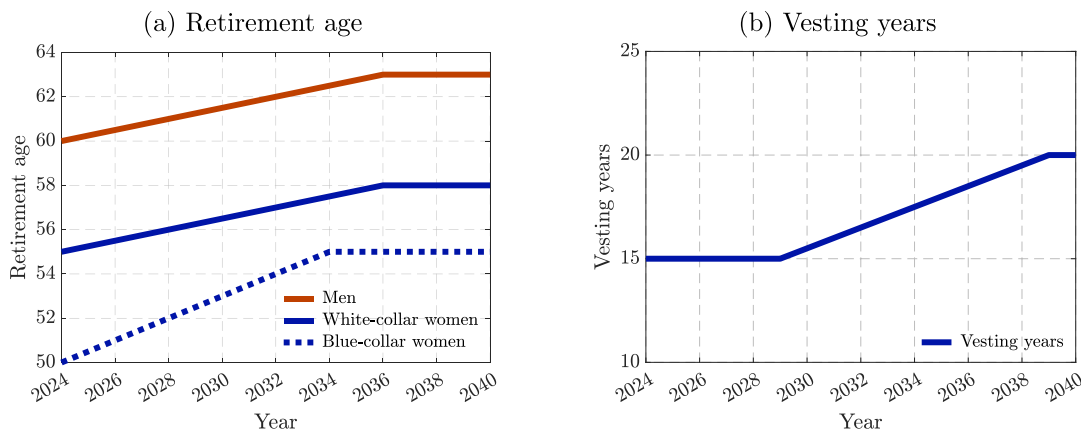


Fig. 3. Retirement age and vesting years under the new policy announced in 2024.

Income and policies are contingent on the formality status of the individuals. The model is estimated to match the empirical data moments as of 2023.

3.1. Simulation of long-run dynamics

In order to project the long-run dynamics of the Chinese economy under different scenarios of policy reforms, we simulate the economy with the parameterization of the benchmark and the evolution of long-run factors. In particular, we feed into the model the long-run trends in three main aspects: (i) demographics, including the population size of cohorts and conditional survival probability by age; (ii) population share by formality; (iii) prices, including the interest rates and wage growth rates. We further assume that the infinitely-lived government has commitment to policies for each cohort, so that each individual acts as if they have perfect foresight of policies throughout their lifetime.

The time-varying factors mean that agents of different cohorts are faced with different parameters to make life-cycle decisions. Given cohort-specific parameters, we solve the life cycle optimization problem for each cohort, simulate the life cycle of each cohort, and aggregate up cohorts coexisting in a given year with proper weights induced by the demographic process to construct time series of variables. The government's pension policies may also change over time, conditional on the policy scenario we consider.

As the problem for an individual is cohort-specific, in the following presentation of the model, we drop the time subscript in the problem and describe the problem from the perspective of a life cycle. For the sake of space, we delineate the environment of the model and delegate

the details of the recursive problem of individuals and the calibration of the model parameters and the corresponding moments to Section A of Online Appendix.

3.2. Demographics

The economy is populated by overlapping generations of individuals. Individuals enter the economy at age 1 and face uncertainty about their lifespan. The conditional probability of survival from age j to age $j + 1$ is denoted ψ_j . The maximum possible age is $j = J$, with $\psi_J = 0$. The size of the cohort is given exogenously. Individuals derive utility from leaving a bequest at death, denoted $\mathcal{V}(\cdot)$. Bequests are assumed to be collected and distributed as a lump sum transfer to the entire population. Individuals enter the economy with no assets except for the transfer of the lump sum bequests.

Upon entering the labor market, an individual is associated with a state of informality $s \in \{1, 2\}$ that is fixed for an individual throughout the life cycle. $s = 1$ refers to the informal sector and $s = 2$ refers to the formal sector. Workers in different sectors differ in their labor productivity and in the public pension system that they enroll. Our choice of modeling formality status as a permanent state is motivated by Chinese data that formality status is tightly linked to one's education and urban household registration status, which makes formality status very predictable and also unlikely to change over the life cycle.⁸ The share of the formal population in a cohort is given exogenously. As will

⁸ Esteban-Pretel and Kitao (2021) use a search and matching framework to model worker transition across the formal and informal sectors over the life

be clear in the projection section, formality has gradually increased to around 50% in 2024 and is projected to continue increasing. To account for this fact, we assume that the formal population share is gradually increasing from older to later cohorts. Over time, older cohorts with a higher informal share exit and younger cohorts with a higher formal share enter; this drives the increase in overall formality.

3.3. Preferences

Without further information on the heterogeneity of preferences between formal and informal workers, we assume that all individuals share the common preference structure and have the same parameters. Individuals maximize the expected utility function over sequences of utility flow $u(\cdot)$ over consumption and leisure $\{c_j, l_j\}_{j=1}^J$ and warm-glow bequest utility $\mathcal{V}(\cdot)$ over assets a_j :

$$\mathbb{E} \left\{ \sum_{j=1}^J \beta^{j-1} \left(\prod_{a=0}^{j-1} (1 - \psi_a) \right) [(1 - \psi_j) u(c_j, l_j) + \psi_j \mathcal{V}(a_j)] \right\}, \quad (1)$$

where β is the discount factor. The expectation is taken with respect to the stochastic processes governing idiosyncratic labor productivity.

We assume that the flow utility takes Cobb–Douglas functional form:

$$u(c, l) = \frac{(c^\gamma l^{1-\gamma})^{1-\sigma}}{1-\sigma}$$

where σ is the risk aversion of households and γ is the utility weight of consumption.

3.4. Incomes, intergenerational transfers, and budget constraint

The markets are incomplete, and there are no state-contingent assets to insure against the idiosyncratic labor income and mortality risks. Individuals can, however, imperfectly self-insure against risks by accumulating one-period risk-free assets. Individuals are not allowed to borrow against future income and transfers, i.e., $a_j \geq 0$ for all j .

Each individual can allocate one unit of disposable time to leisure (l) or market work (h). Individuals' earnings are given by $y_w = \tilde{w}h$, where $\tilde{w}$ denotes the wage rate per work hour h for each individual and is given as

$$\log \tilde{w} = \omega + \theta_j^A + \theta_s^S + \theta_{ij}^E,$$

where ω is a general equilibrium wage component, θ_j^A is the age component, θ_s^S is sector labor productivity, and θ_{ij}^E is an idiosyncratic wage income shock that evolves stochastically. The wage income is therefore given as $y_w = \tilde{w}h$.

Adult children in China — as well as in many other developing countries — are supposed to provide financial support to their parents. As the private transfers constitute a major part of old-age support in particular for the rural population who have low public pension benefits, we explicit model the intergenerational monetary transfers to account for the potential role of private transfers for policy effects. We denote the transfers made by an agent to her parent generation as y_m and the transfers received from her child generation as y_r .

As we do not explicitly model the fertility process, therefore, the identity of parent and children generations, we assume that agents are children of the generation born 25 years ahead of them. Agents make transfers to their parents once they enter the labor market (age 20) and until the parent generation completely exits the economy (age 75). Agents receive transfers starting from age 45 — when their children generation enter the economy — to the end of their life. The size of the transfers made is proportional to their income, which is affected by the age and formality status of the agent. The size of transfers received

is computed by dividing the total amount of transfers made by their children generation by the size of the cohort.

Assets evolve according to the law of motion:

$$a' + c = a(1 + r) + y_w + y_b + y_r - y_m - \mathbb{T} \quad (2)$$

where y_b denotes public pension income and $\mathbb{T}$ denotes the sum of all taxes. Individuals are not allowed to borrow against future income and transfers, i.e., $a_j \geq 0$ for all j .

3.5. Public pension system

To model the two-tier pension system in China, we assume that the government operates a pay-as-you-go pension system that covers only formal workers and an old-age pension system that covers informal workers.

Formal sector. A proportional public pension contribution tax τ_{sl} is imposed on the earnings of formal workers until age $J_R - 1$. Agents start to receive public pension benefits once they reach retirement age J_R .

To approximate the Chinese pension system, we introduce the cumulative pension contribution b as a state variable to the problem of the household. b evolves over the life cycle and determines the level of pension benefits after retirement. In every working period, the existing cumulative contribution stock grows at the market deposit rate r and the formal workers make contributions that are τ_{sl} fraction of wage income, i.e.,

$$b_j = (1 + r)b_{j-1} + \tau_{sl}y_w. \quad (3)$$

The pension benefit that an individual receives is contingent on contribution b_{J_R} , i.e., total pension contributions made up till age J_R . Under the current rule, contributions will be deposited into a social-pooling and an individual account by a fixed fraction, and the benefit is calculated as the sum of withdrawals from the two accounts. The social pooling account is redistributive and the individual account is only tied to the total contribution made. To capture the fact that the current pension system depends on both the contribution years and the total contribution and is redistributive in nature, we approximate the benefit in the first year of retirement as

$$y_b = \beta_1 J_R + \beta_2 b_{J_R} \quad (4)$$

With specification (4), we implicitly assume that the social pooling and individual accounts have the same internal rate of returns.

Based on the formula of pension benefits, pension income is effectively fully indexed to price changes but only partially indexed to wage growth. For current cohorts, we assume that the benefit level increases by 2% per year in addition to inflation, which is consistent with evidence of Fig. 2. We further assume that the growth rate gradually decreases to 1% in the long run, due to the slowdown of aggregate wage growth. We calibrate the parameters β_1 and β_2 to match the average and the interquartile gap of the public pension benefits paid in the year 2023.

As there is no ex-ante heterogeneity for agents within the formal sector, we assume a common retirement age for all formal workers. We set it to be age 56, which is the average effective retirement age before 2023.⁹ For simplicity, we assume that formal workers retire once they reach the pension eligibility age and there is no recall after retirement.

⁹ Apparently, age 56 is calculated by pooling population who are eligible for different retirement ages (50, 55, or 60 for policy before 2023). See Gao (2025) for a framework that evaluates the effects of the policy for individuals with different characteristics.

cycle. Gao and McKiernan (2023) further endogenize job arrival rates with a firm vacancy posting problem.

Informal sector. We assume that informal workers start collecting pension benefits at the same age at a fixed age of 60. The informal workers may continue to work while collecting the old age pension benefits. The maximum age at which an informal worker can work is 65 years and is independent of their pension eligibility age. The informal worker can change their labor supply at older ages in response to changes in their pension policy. This allows us to study the response of informal workers' labor supply at older ages to policy reforms.

Under current policy, informal workers are only eligible for old-age assistance programs that have much lower benefit levels compared with their formal peers. We assume that informal workers contribute 2.5% of their wage income to the RURPS before retirement and receive pension benefits that are 6% of the average pension income of EBPS pension recipients. These numbers reproduce in the model the average contribution for RURPS participants before retirement and the average benefit level after retirement in 2023 as in Table 1.

3.6. Fiscal policy

The government spends an exogenous amount of G on public purchases of goods and services and issues one-period riskless debt D' , which pays the annual interest rate of r . In addition to payroll pension contribution taxes, revenues are raised from taxation on wage income, capital income, and consumption. The wage income tax schedule $\tau_w(\cdot)$ is progressive and is applied to all workers. Capital income and consumption are taxed at proportional rates, each denoted as τ_a and τ_c , respectively. The government budget constraint is satisfied for every period:

$$\sum_{\chi} [\tau_{sl} y_w(\chi) + \mathcal{T}_l(y_l(\chi))] \mathbb{1}_{[s=2]} \mu(\chi) + \sum_{\chi} [r a(\chi) \tau_a + \tau_c c(\chi)] \mu(\chi) + D' = G + \sum_{\chi} w_r(\chi) \mu(\chi) + (1+r)D \quad (5)$$

where the taxable income $y_l(\chi) = y_w(\chi) - \tau_{sl} y_w(\chi)$ for formal workers captures the fact that the pension contribution is tax deductible, and $y_l(\chi) = 0$ for informal workers, i.e., informal workers do not pay personal income taxes. This choice of modeling is motivated by the fact that the personal income tax in China constitutes a very small source of government tax revenues.¹⁰

3.7. Parameterization and model fit

We set the risk aversion parameter $\sigma = 4$ and set the discount factor $\beta = 0.96$ such that the capital-to-output ratio is 1.6 as in the data. We set $\gamma = 0.4$ to match the mean work hours. As in Choukhmane et al. (2023) and Banerjee et al. (2023), we assume that transfers made to their parents amount to 9% of income for both formal and informal workers.

The statutory contribution is 24% of the wage, based on the social pooling 16% and the individual accounts 8%. However, since self-employed members only contribute 12% to social pooling accounts, and compliance is significantly below perfect, we calibrate the effective contribution as 21% of wage income that reproduces the national account statistics reported in Table 1.¹¹ The current pension benefit in

terms of the replacement rate at retirement is 50% of the average wage; the pension is indexed to the real wage growth, which is 2%.

In Table 2, we present the fit of the model in the cross section of 2023. As the upper panel of Table 2 shows, our model provides good fitness of the levels of earnings, consumption, pension contribution, and benefits. In the lower panel, we further present distribution moments, including the standard deviation of log earnings and wages, and the mean household net savings rate. In the data, the overall dispersion of earnings and wages is larger than that by group, which is also well captured by our model. Our model correctly predicts that the savings rate for informal workers is lower than that of formal workers, which is consistent with the data. However, the level is lower than that in the data. This suggests that our model may miss some factors that affect the incentive to save for Chinese households.¹² Despite its vital importance, explaining the savings puzzle of Chinese households by projecting the long-run trend of asset prices is challenging and far beyond the scope of our studies. As an alternative approach, we perform sensitivity checks with alternative values of preference parameters to investigate the role of the savings incentive in the following sections.

4. Long-run projections of policy reforms

To project the trends of variables from 2024 to 2100, we consider 157 cohorts, starting from the cohort entering the labor market in year 1944 (born in 1924) and ending the life cycle in year 2024, to the cohort entering the labor market in year 2100. In 2021, the population of China reached its historical peak of 1.412 billion. We only model the population above the age of 20, which is around 76% of the total population (1.08 billion).

4.1. Inputs of long-run factors

We take the projection of the population size by age for each cohort directly from the World Population Prospects 2024.¹³ We use the projection in the medium fertility scenario as our baseline and check the robustness of the results in other scenarios.¹⁴ Fertility rates from 1950 to 2023 are taken directly from data. Recent years have seen a historically low fertility rate, hitting less than 0.9 births per woman. The WPP projects that the fertility rate will gradually recover to 1.35 by 2150.

Motivated by the fact that the formality share is largely affected by the urbanization process, we assume that the formality share by region is a constant over time and project the urban population share over time using experience from countries with comparable backgrounds. As shown in Fig. 4(d), the share of urban population in 2023 was 64%. It is projected to rise to 80% in 2045 and further to 90% in 2072. Assuming that 80% of the urban population is formal and 30% of the rural population is formal, we obtain that the share of formal employees in the total working population is projected to reach 70% by 2050 and stabilize at the level of 80% after 2075.

The real wage growth up to 2023 is extracted from the National Bureau of Statistics website and is projected to be reduced to 2% by 2050 and then to remain constant thereafter. In Section B of the

¹⁰ According to the State Administration of Taxation of the People's Republic of China, over 70% of Chinese wage earners are exempted from individual taxes, and only around 10% of the wage earners pay income taxes above the 3% marginal rate. See <https://www.chinatax.gov.cn/chinatax/n810219/n810780/c5235243/content.html> for more details.

¹¹ According to The Report on the Inspection of the Implementation of the Social Insurance Law of the People's Republic of China by The National People's Congress Standing Committee Law Enforcement Inspection Group, The compliance rate of employee pension insurance (the proportion of contributors to insured persons) was 80.8% in 2022.

¹² For example, we only consider the general form of asset holding, while the housing boom in China plays a key role in shaping the rapidly increasing trend of the household savings rate, particularly in the urban sector (Chamon and Prasad, 2010; Wang and Wen, 2012; Chen and Wen, 2017).

¹³ Data can be found at <https://population.un.org/wpp/>.

¹⁴ According to definition of WPP, "the medium scenario projection corresponds to the mean fertility and mortality and median net migration of several thousand distinct trajectories of each demographic component derived using the probabilistic model of the variability in changes over time. Prediction intervals reflect the spread in the distribution of outcomes across the projected trajectories and thus provide an assessment of the uncertainty inherent in the medium scenario projection".

Table 2
Model fitness: Cross-section of 2023.

	Data			Model		
	All	Informal	Formal	All	Informal	Formal
Annual earnings per worker, CNY	69,429	51,875	88,524	68,042	46,373	90,490
Consumption per worker, CNY		18,175	32,994		27,100	38,444
Pension contribution per worker, CNY		1660	18,590		1325	19,003
Pension benefit per retiree, CNY		2671	44,912		2645	44,777
S.d. of log wage	0.71	0.65	0.58	0.41	0.33	0.30
Net saving rate	0.30	0.16	0.36	0.14	0.10	0.16

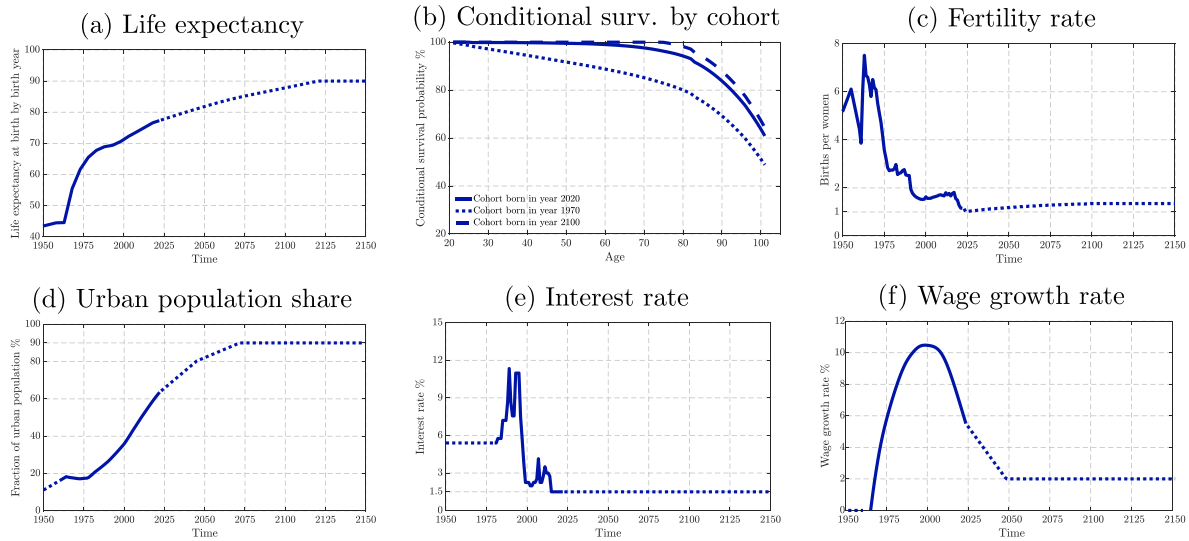


Fig. 4. Model inputs for projection.

Online Appendix, we provide more details on the data sources and the imputation and forecasting of these variables.

In the following, we start with the scenario before the newly-announced reform in October 2024, and then examine the effects of the reform by comparing the outcomes under the new policy with those under the scenario that the old policy is continued into the future. Next, we examine the effects of several policy proposals that aim to achieve better fiscal sustainability and reduce inequality among formal and informal workers. In all experiments, we study the impact of policy on household welfare and household behaviors, including consumption, savings, and labor supply, and the impact on the government fiscal budget.

4.2. Effects of the 2024 reform

Since we treat the average household member as the unit of economic analysis in the model, we take the average retirement age proposed by 2024 over men and women to approximate the effect of the reform. Fig. 5 plots the change in retirement age over time in the model.¹⁵ Also note that our formulation (4) captures the policy effect that the annual pension income increases as agents' contribution years increase.

Not surprisingly, the increase in retirement age increases the size of the labor force, as shown by Fig. 6(a). By 2037, the labor force is projected to increase by 40 million and remain stable in the next couple of decades. The change in retirement age reduces the dependency ratio from around 1.6 to 1.5 in 2050, as shown in Fig. 6(b).

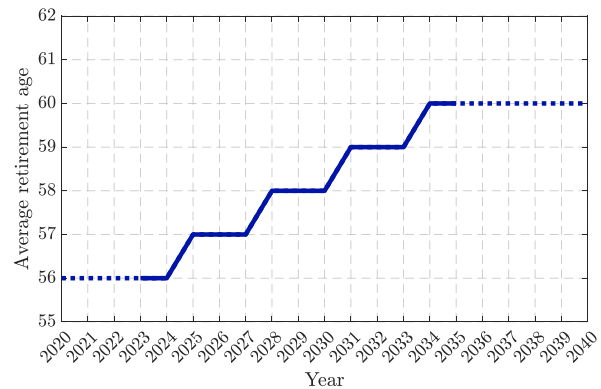


Fig. 5. Retirement age in the model.

In Fig. 7, we plot the net public pension budget in the current year and in cumulative terms up to 2065. In both panels, the blue solid lines refer to the projection under the scenario of continuing the policy before the 2024 reform, and the red solid lines refer to the projection under the 2024 reform.

As is consistent with the data, the Chinese public pension budget still ran slightly a net surplus in 2023. As our projection shows, it quickly runs into a net deficit by 2025. By 2065, the net pension deficit of the pre-reform scenario is projected to reach approximately 32 trillion CNY in 2023 prices, equivalent to 12% of GDP in that year. We compute the discounted present value of debt using the market deposit rate, the interest rate given in Fig. 4(e). The cumulative net deficit of the public pension system is expected to be 110% of GDP in 2023 by 2050 and 270% of GDP in 2023 by 2065.

¹⁵ As in the benchmark policy, we compute the average retirement of the population. In reality, the average retirement age, which does not necessarily take an integer value, will increase year by year. However, since our model period is a year, we can only approximate the gradual increase as a step function.

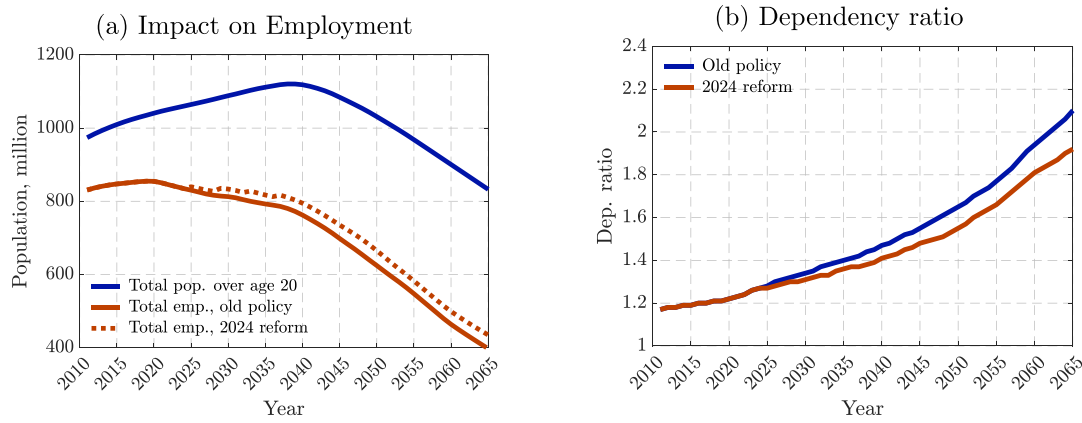


Fig. 6. Impact of the 2024 reform on the size of labor force.

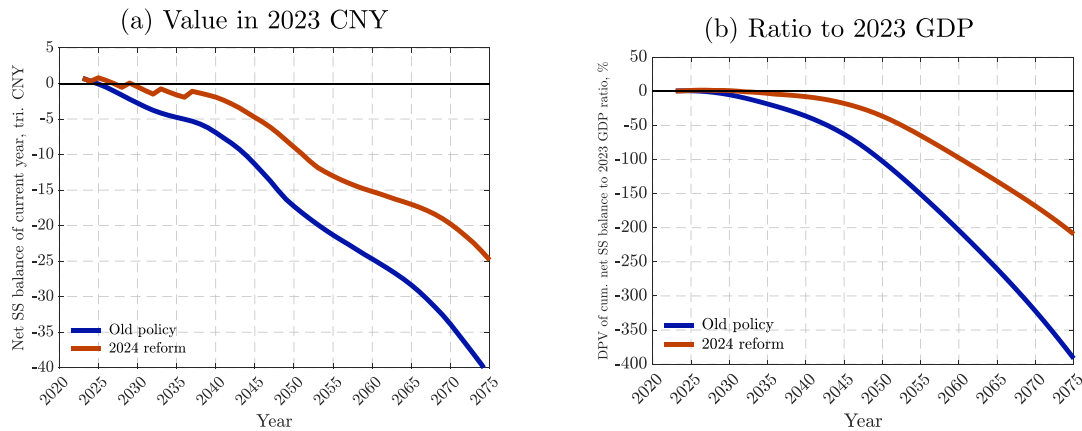


Fig. 7. Impact of the 2024 reform on government revenue.

Note: The cumulative present value of public pension deficits is discounted using the market deposit rate.

Despite the increase in annual pension income after the reform, the reform greatly reduces the financial burden, as the overall effect is dominated by reduced public pension reception years. As the red line in Fig. 7 clearly shows, the 2024 reform can prolong the net non-deficit period into the 2030s. Furthermore, the 2024 reform will reduce the annual public pension deficits from 12% of GDP to 8% and the cumulative discounted present value of deficits by around half by 2065.

One may wonder what contribution rate can prevent the government from running a deficit in the long run. We find that under our assumption of the benchmark scenario, the government has to impose a 36% contribution rate to maintain “revenue balancing” of the pension system over the long run, i.e., up to 2100. Given that the direction of contribution rate adjustments in the past decade has been largely downward, this seems both unlikely and undesirable from a labor market perspective, both in terms of the global competitiveness of Chinese urban workers and in terms of the extent of distortions between factors of production within China.

Through the lens of the model, the aggregate savings rate will respond to all three trends—the demographic, employment, and growth trends if the old policy is continued. First, as the formal population share continues to increase, it can mechanically decrease the aggregate savings rate, as formal workers have a lower savings rate relative to their informal peers, since they have better social insurance. Second, the ageing process may lead to mechanical shifts of population composition and behavioral responses through life cycle re-optimization. On the one hand, the aging process increases the share of retired in the population. This tends to decrease the savings rate, as retirees are net negative savers. On the other hand, given a fixed retirement age, the

ageing process means each individual has longer retirement periods to finance, which will incentivize agents to save more in their early life. Third, the slowdown of wage growth, as reflected by the changing age profile for different cohorts, will affect agents’ incentive to save as well.

The impact on the aggregate savings rate is apparently the result of all these factors. As the blue line in Fig. 8(a) shows, the savings rate increases slightly over the next 15 to 20 years, mainly due to a decrease in aggregate growth. In the long run, the savings rate decreases as the increase in the share of the formal and retired population dominates.

We further investigate how the 2024 reform may affect the savings rate. The 2024 reform may affect the formal workers’ savings incentive through three channels. First, it increases the ratio of working time relative to retirement time for an individual, or alternatively, it increases the ratio of the employed population to the retired population, which increases the aggregate savings rate. Second, since each individual expects to work for longer periods to accumulate enough assets for retirement, they may save less in each period. Third, though the benefit rule adjusts for longer contribution years, it on net is still a decrease in pension wealth. Using several earlier reforms of the Chinese pension system as quasi-experiments, Feng et al. (2011) and He et al. (2019) find that a decrease in pension wealth has led to more household savings and labor supply. As the red line in Fig. 8(a) shows, relative to the pre-reform scenario, the savings rate slightly decreases in the long run.

In addition, we examine the trend of aggregate working hours under the scenario of old policy and the 2024 reform. In Fig. 8(b), we report the ratio of aggregate labor supply under the 2024 reform to that under the case without reform. As Fig. 8(b) shows, the aggregate

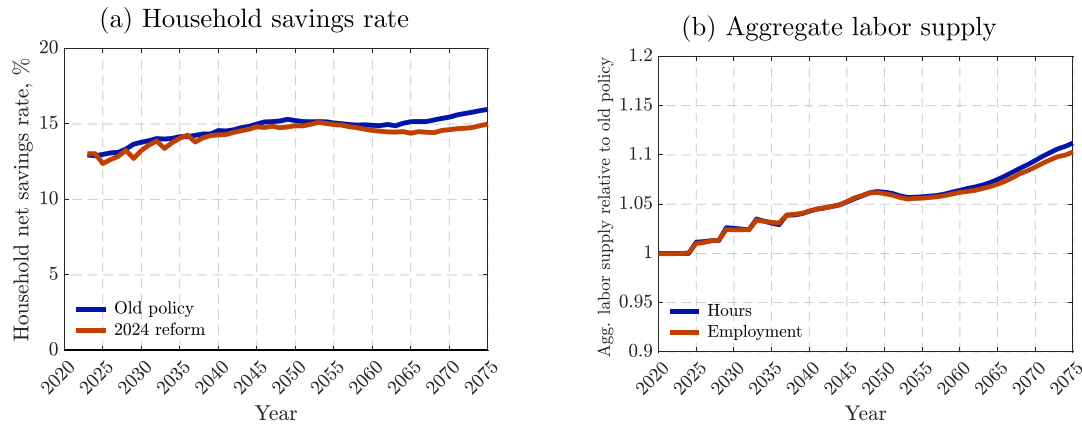


Fig. 8. Impact of the 2024 reform on household savings rate and aggregate labor supply.

working hours increase by up to 6% by 2050, and the increase is mainly explained by an increase in employment (the extensive margin) as shown in Fig. 6(a).¹⁶ Fig. 8(b) also shows that the reform decreases the average working hours despite increasing total employment, as shown in Fig. 6(a). This is also mechanical, as the delaying retirement policy increases the amount of older workers, who are associated with lower hours. The trend of total working hours is driven by the extensive margin, that is, the increase in labor force participation.

5. Further reform options

To address the main challenges documented in Section 2, namely, fiscal unsustainability due to ageing and the wide gap between formal and informal workers, we evaluate several potential follow-up policy proposals. The first proposal aims to enhance fiscal sustainability of the formal pension system, therefore only involving the change of policies for formal workers. The last aims to alleviate disparity by formality and the last proposal aims to address both inequality and unsustainability issues.

For each policy proposal, we consider two ways of implementation. First, we simply change the public pension policy without rebalancing the government budget using other policy tools. This gives us a sense of how the policy would affect the fiscal budget on its own. Second, we ensure that the government budget is rebalanced in the long run with either lump-sum transfers or tax reimbursement to examine the welfare effects of the policy reforms. The reference point for the government balance is the accumulated government debt by the year 2100 under the scenario that the 2024 new reform is implemented.

For each reform, we assume that the reform starts to take place from 2025 and that the government has a commitment to the pension wealth accumulated before 2025. To be more specific, cohorts that have retired before 2025 collect pension benefits with the rule under the 2024 reform; cohorts that have already started working, but not yet retired, have two phases of pension wealth accumulation. Therefore, each cohort may be affected by the policy reform to a different extent, and it takes full working years of a life cycle to see the cohort that is covered by the new policy over his/her all life.

5.1. Reform options

1. **Alternative indexation of pensions in payment** In the first scenario, we propose further altering the indexation of pension benefits from the baseline, which is partially indexed to real

wage growth, to a one that is indexed to CPI only. The current policy guidelines allow indexation based on a range between price, wage, and GDP growth. The CPI indexation usually represents a minimum indexation scenario, which shall significantly reduce the benefits when the retirees age. In the model, we simply set the agent's pension income to be fixed at the level given by Eq. (4).

2. **Increasing level of residents' pension** To address the pension inequality between formal and informal workers, we evaluate how increasing the level of benefits for informal workers can add to the financial burden of fiscal sustainability. Earlier studies, such as Jung and Tran (2012), show that extending public pension coverage to informal workers leads to welfare gains by providing better insurance to disadvantaged groups despite potential efficiency losses. To proceed, we set the contribution to be the same as in the baseline but increase the replacement ratio from the current level of 6% of the formal pension benefit to 30%, or equivalently 2% of GDP per capita to 15% of GDP per capita, which is more consistent with the prevailing international practice (Chomik et al., 2024). Increasing pension benefits can potentially lead to better outcomes for the informal population in many other dimensions, such as better health, as shown by Huang and Zhang (2021). In this paper, we limit our exploration only up to consumption and labor supply of households and leave the exploration on health consequences for future research.¹⁷ In practice, the government can consider richer way of implementing the policy such as means-tested pension benefits as considered in Wheadon et al. (2024).
3. **Consumption tax based financing** The final option that we model is to replace the contribution of the effective 21% of pre-tax wage with a post-tax consumption contribution rate of $\tau_{sc} = 25\%$. Similar proposals have been made in countries such as Mexico (Hernández et al., 2017) and Germany (Ruppert et al., 2024). By using consumption tax instead of a payroll tax, the link between formality and public pension contribution is weakened, in the sense that the consumption tax-based public pension system is able to cover the entire population regardless of their formality status. Furthermore, we assume that the old age resident pension for the informal sector gradually phases out as the consumption tax gradually phases in. We further assume that all contributions go to an individual account but up to a notional base without considering the fiscal

¹⁶ The average hours change to a small extent, slightly increasing in the short run but eventually decreasing in the long run, mainly driven by the compositional effect that the share of informal workers decreases.

¹⁷ To evaluate the welfare consequences through health, one will need a framework as in Deng et al. (2023).

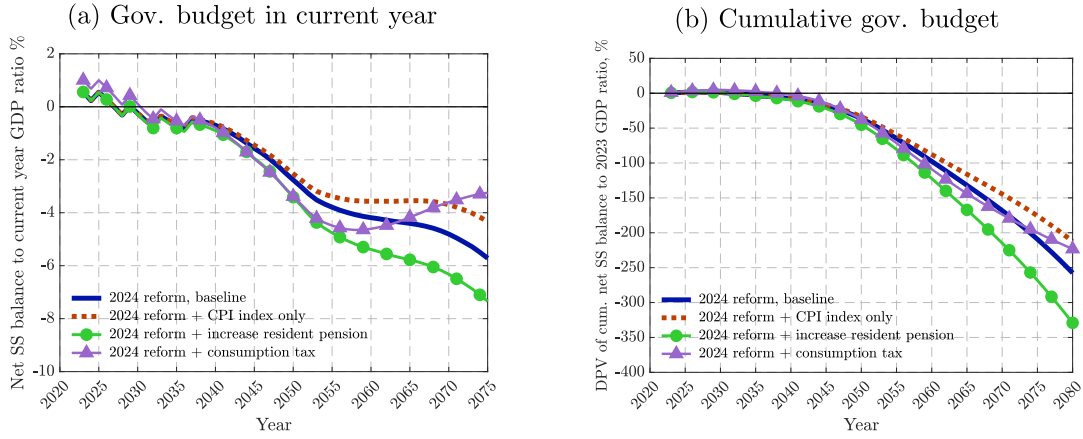


Fig. 9. Impact of further reform options on government revenue, ratio to 2023 GDP.

transition cost.¹⁸ In the model, the state variable of pension contribution then evolves following

$$b_j = (1 + r)b_{j-1} + \tau_{sc}c. \quad (6)$$

By assuming that the return is the same as the average return under the current policy, we obtain the formula for benefit level as

$$y_b = \tilde{\beta}_2 b_{JR} \quad (7)$$

where $\tilde{\beta}_2$ is the factor that matches the mean benefit level given the total contribution in the data of 2023. Relative to (4), specification (7) apparently does not feature redistribution.

5.2. Impact on government budget

First, we examine the impact of policy reforms on the government budget. In evaluating the impact of the policies, we assume that there are no accompanying tax or transfer policies. We plot the impacts on the government budget in the current year as a share of GDP in Fig. 9(a) and on the cumulative government budget in Fig. 9(b). In both panels, the benchmark scenario without further reforms is plotted in solid blue lines.

We first evaluate how the proposed policy change would affect the fiscal budget without accommodation of other policy changes. As the dashed lines in Fig. 9(a) show, indexing the benefits to the CPI only reduces the deficits to 4% of GDP by 2075. In terms of the accumulated government budget, it reduces the fiscal deficit from 250% of GDP to 200% in 2065. As the green lines with circle markers in Fig. 9 show, increasing the level of resident pension benefit for informal workers from 6% of the average formal pension income to 30% of the average formal pension income will increase the public finance deficit to 7% of GDP in 2075 at prices of 2023, with accumulated debt reaching 320% of GDP of 2023 compared to 250% in the baseline scenario. However, due to the foreseeable decrease in the share of informal workers, further increasing the benefits for informal workers does not add too much of a fiscal financial burden. The consumption tax case is trickier, as it involves both a decrease in the contribution and the payout of the pension benefits. As a result, in the first two decades of consumption tax-based contribution, it exhibits a deficit similar to the 2024 reform scenario in the first stage, as existing pensioners

still receive high benefits, as the purple lines with triangle markers Fig. 9 show. However, by around 2065, the accumulated reduction in expenditures exceeds the revenue gap compared to the normal wage-based contribution scheme, leading to a smaller budget deficit. By 2075, the consumption tax reduces the annual government deficit to less than 4% of GDP and the cumulative deficit would drop to 210% of GDP in 2023.

5.3. Impact on welfare

Welfare calculation. In the previous subsection, we focus on the impact of alternative reform schemes on the government budget by being agnostic about how the government raises the funds to finance the extra deficits or spend the extra revenue. In order to compare welfare across the schemes, we need to hold the government budget fixed and make assumptions on how the extra revenues are raised/distributed.

We assume that the government can use lump-sum transfers or adjust the pension contribution rate to maintain the same deficit level as under the policy environment of the 2024 reform, i.e., the deficit of government fiscal revenue to GDP ratio is 300% by year 2100. We further assume that the size of the lump-sum transfer is indexed by the average wage level by year and will be applied to all individuals regardless of their age and formality status. This effectively introduces redistribution across the formal and informal workers as they have different income levels. In addition, it also leads to redistribution for individuals at different stages of life. Another way of rebalancing the long-run government budget is to change the contribution rate, which is easier to implement compared to lump-sum transfer in practice, but is not compatible with all reform options. For example, when we want to replace the payroll tax with the consumption tax (option 3). Therefore, we consider lump-sum transfer as the benchmark implementation for all cases and consider alternative implementation of changing contribution rate for options 1 and 2. The choice of lump-sum taxes/transfers is intended for ease of interpretation and no distortion to individuals' decision-making. Apparently, a more complicated way of controlling the aggregate fiscal budget under alternative assumptions, such as taxing or redistributing to specific households, may produce different outcomes. The lump-sum tax/transfer approach serves as a simplified basis for comparison.

The government would run a surplus relative to the benchmark in several cases. When the pension benefits are indexed to inflation only, the government would reimburse households with annually 3098 CNY at the wage level of year 2023. Alternatively, the government can reduce the pension contribution rate from 21% to 13.5%. The government runs a deficit if the replacement rate of informal workers' benefits is raised. In this case, the government imposes a 1269 CNY

¹⁸ We only analyze the economic impact of the policy reform with the assumption of full commitment and perfect information. Further extension to consider behavioral responses subject to information frictions in implementation such a policy, though, is beyond the scope of our analysis.

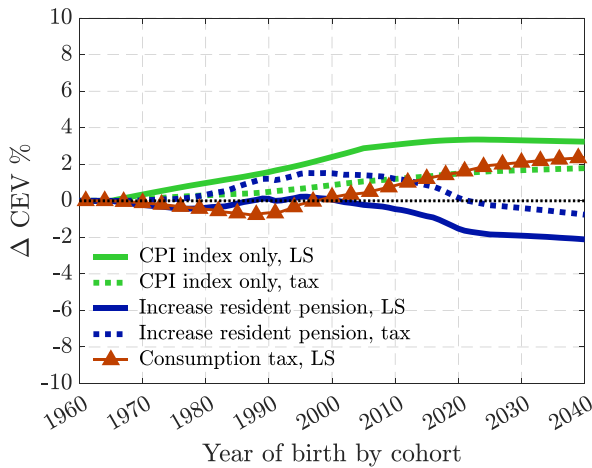


Fig. 10. Overall welfare change under further reform options.

lump sum tax on all households at the wage level of year 2023. For more details on the compensation that government needs to make to maintain government budget position as in the benchmark by 2100, see Section C of the Online Appendix.

Following Conesa et al. (2009), we measure welfare using *Consumption Equivalent Variation* (CEV), which quantifies changes in average lifetime utility as the percentage change in consumption needed to make individuals indifferent between a baseline policy and an alternative policy scenario. To be more specific, given the form of the utility function, the welfare consequences of switching from a steady-state consumption-labor allocation (c_0, l_0) to (c_*, l_*) are given by

$$CEV = \left[\frac{W(c_*, l_*)}{W(c_0, l_0)} \right]^{\frac{1}{1-\sigma}} - 1$$

where $W(c, l)$ is the expected lifetime utility at birth of a household, given a policy system.

The aggregate social welfare is assumed to be ex ante utilitarian utility that weights each individual equally, in spite of their formality status, i.e.,

$$SWF(\mathcal{G}) = \int V(a=0, s, j=1|\mathcal{G}) d\mu(\mathcal{G}),$$

where $\mathcal{G}$ refers to a set of government public pension and lump-sum tax/transfer policies, μ is the distribution of agents, and the value function $V(a=0, s, j=1|\mathcal{G})$ refers to agents' ex-ante utility when they enter the labor market. Using a similar way, we can also define welfare change for a permanent group—in our specification, the formal and informal sectors, with the CEV of the ex-ante utility.

Results. Fig. 10 presents the overall welfare changes. The x-axis lists the cohorts by their year of birth, and the y-axis represents the changes in welfare for each cohort. As Fig. 10 demonstrates, for cohorts born after 2000, the scenario of CPI indexation only reimbursed with lump-sum transfers provides the most substantial welfare gains—amounting to 4% increase in annual consumption. The increase in the informal replacement rate or the use of the consumption tax also records a welfare gain of 2% CEV in the short run. For future cohorts born after 2010, while all other scenarios present flat welfare gains, the gains from increasing resident pension option diminish to slightly negative for the cohort born in 2040.

As policy reforms may affect the formal and informal sectors differently, in Fig. 11, we report welfare changes for the formal and informal sectors using the same assumptions as in the construction of Fig. 10. Also note that the weights of the aggregate welfare function over the two sectors are changing by cohort based on our projection of the formalization process, where the formal sector is expected to grow rapidly in the next decades, but stabilize in the long run.

As the solid lines with triangle markers in Fig. 11(a) show, indexing pension benefits to the CPI only leads to welfare gains for both the informal and formal sectors, with an overall effect of approximately 2% for the cohort born in 1980 and around 4% for the cohort born in 2040. Not surprisingly, since the lump-sum transfer is independent of workers' formality status, the informal workers experience a more substantial welfare gain relative to their formal peers. However, this policy also leads to better outcomes for the formal population in spite of a loss of pension wealth. This is because a lump-sum transfer at younger ages helps agents make better intertemporal decisions over the life cycle. When agents are young, they are subject to liquidity constraint; the lump-sum transfer received at younger ages allows individuals to increase consumption earlier on. In this policy scenario, the welfare gain is reaching 4.5% for the informal population in the long run, while the gain for the formal sector is 3%.

If, instead, a lower contribution rate is implemented, the overall welfare impact is less. As the dotted lines in Fig. 11(a) show, a lower contribution rate would only affect formal workers, while keeping informal workers unchanged, thus not helping to improve the redistribution between formal and informal workers. Therefore, the overall welfare gain is smaller than balancing government budget with lump-sum transfers, although the formal population has similar gains.

An adequate resident pension is a common option to reduce the income inequality of the elderly (Lu et al., 2014). Increasing the social replacement rate for informal pensions, by nature, leads to welfare gains for the informal population. However, the welfare gain for the informal population comes at the expense of reduced welfare for the formal sector, as the government will need to use a lump-sum tax or increase the contribution rate for formal workers to finance the increase in resident pension. Since the lump-sum transfer is imposed on the whole population while the contribution rate applies only to the formal workers, the welfare gain for informal workers is larger when it is implemented with government adjusting the contribution rate, as the dotted green line shows. As the green lines in Fig. 11(b) show, apparently, the welfare gain for informal workers increases the longer they experience the expansion of pension benefits, making the welfare gain greater for later cohorts. When implemented with a lump-sum transfer to rebalance government budget, increasing the informal pension benefit replacement rate to 30% of the formal pension level results in a welfare gain of as high as 3.5% CEV for the informal sector in the long run, although this comes at the expense of a welfare loss of almost the same size for formal workers in the same cohort. However, with adjustment of the contribution rate, the welfare gain for informal workers amounts to 7.5% of CEV, while the welfare loss to formal workers is only 3% of CEV, leading to smaller overall welfare losses when evaluated as a whole. As the share of the formal population continues to increase, the aggregate welfare gain from increasing benefits for the informal becomes less, but the costs to finance the increase in resident pension also decrease per formal worker. Therefore, we expect the welfare change to remain slightly negative in the long run.

Fig. 11(c) evaluates the welfare effects under the policy scenario of using consumption tax-based financing. The welfare change for this scenario is not monotone as we see in other scenarios. Despite the fact that it can distort intertemporal consumption-savings decisions of individuals, the consumption tax at 25% is a reduction in the overall burden of the pension contribution tax. As the consumption tax is uniform, it is a relaxation of the pension contribution tax burden for the formal sector, while it increases the tax burden for informal workers. In addition, consumption tax-based financing is implemented with a lump-sum transfer. Therefore, the formal sector experiences increasing welfare gains until the cohort born around 1995, reaching 4.7%, after which it drops to a flat 3.8% for cohorts born after 2020. In contrast, the informal sector faces initial welfare loss until those born after the mid-1980s, the welfare loss gets smaller for the following cohorts and stays flat at about 2% welfare loss for those born after 2020.

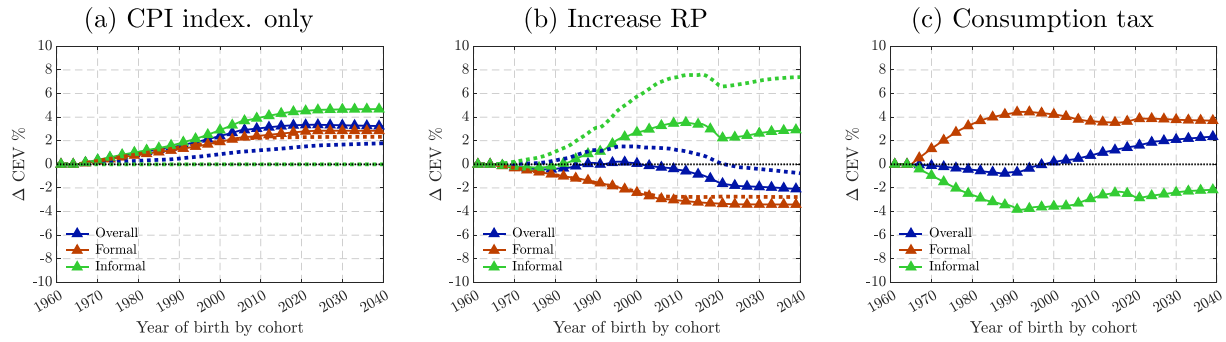


Fig. 11. Welfare change by formality.

Note: In panels (a) and (b), results under the assumption of government reimbursement in the form of lump-sum transfer are depicted in solid lines, and results under government tax reimbursement are depicted in dotted lines.

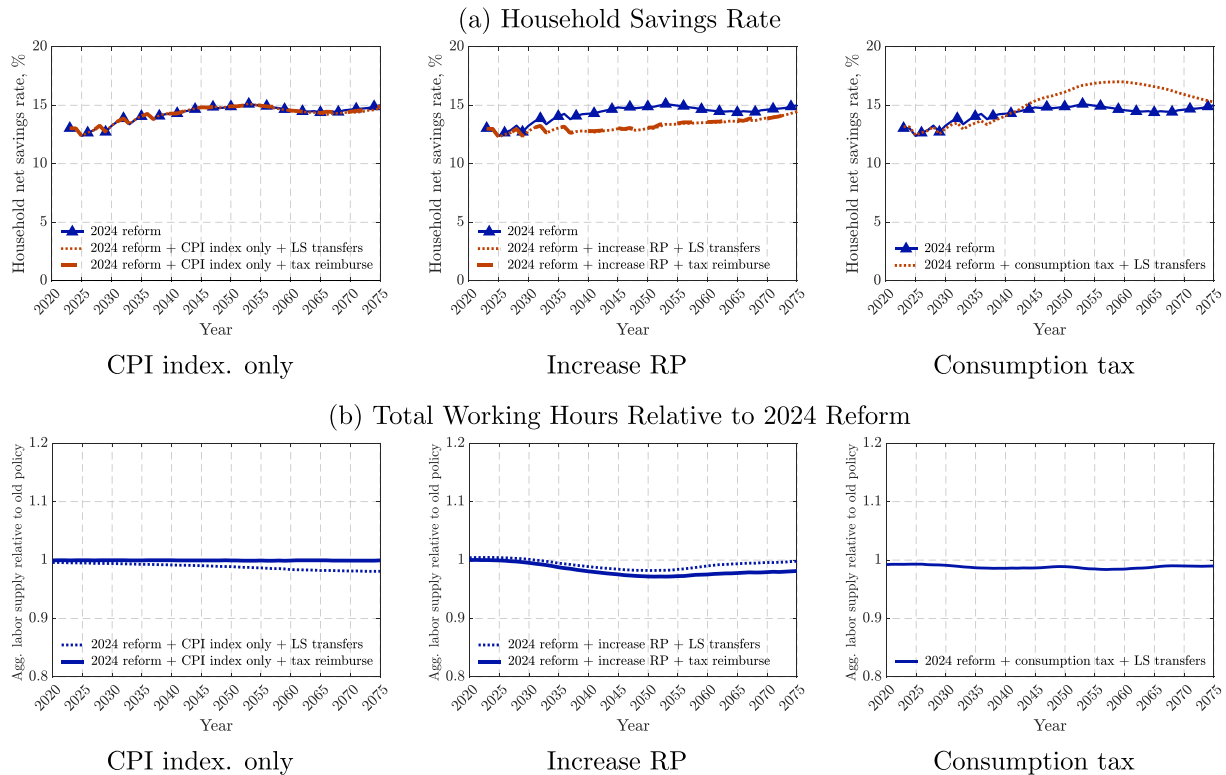


Fig. 12. Impact of reforms on households savings and labor supply.

5.4. Impact on household decisions

In Fig. 12, we report the impacts of policy reforms on household behavioral responses in terms of their savings and labor supply. For each policy, we plot both the impacts under scenarios with and without a lump-sum transfer/tax.

As column 1 in panels (a) and (b) shows, removing the partial indexation of wage growth is a decrease in pension wealth for formal workers and an increase in incomes earlier on. Therefore, formal workers would want to increase their labor supply and savings to insure themselves against the negative wealth effect. However, if lump-sum transfer is introduced, labor supply decreases. The labor supply does not decrease if a lower contribution rate is implemented. Informal workers experience a pure wealth effect, while formal workers achieve better intertemporal allocation with more transfer at younger ages. Overall, the impact on the household savings rate is negligible.

In column 2 of panels (a) and (b), we examine the impact when the government increases the pension benefits for informal workers.

The policy reform achieves better reallocation across formal and informal workers, but will affect formal and informal workers' behaviors differently. The impact on savings is dominated by the effect that the informal workers are better insured with public pension therefore reduce their savings. This is also evident as the impact on the savings rate in the long run diminishes. Furthermore, the wealth effect also incentivizes informal workers to work less. If the government finances the increase in resident pension with an increase in contribution rate, formal workers also reduce labor supply. If instead a lump-sum tax is imposed, formal workers are incentivized to increase labor supply, which leads to a lower decrease in aggregate labor supply.

Finally, in column 3, we examine the impact of consumption-tax-based financing. The consumption tax would alter agents' intertemporal decisions by encouraging them to consume less and save more. However, this only happens to formal workers. After the reform, the informal population, who have a lower disposable income and decrease their savings. As column 3 in panel (b) shows, the net household savings rate increases as the impact on the formal sector dominates.

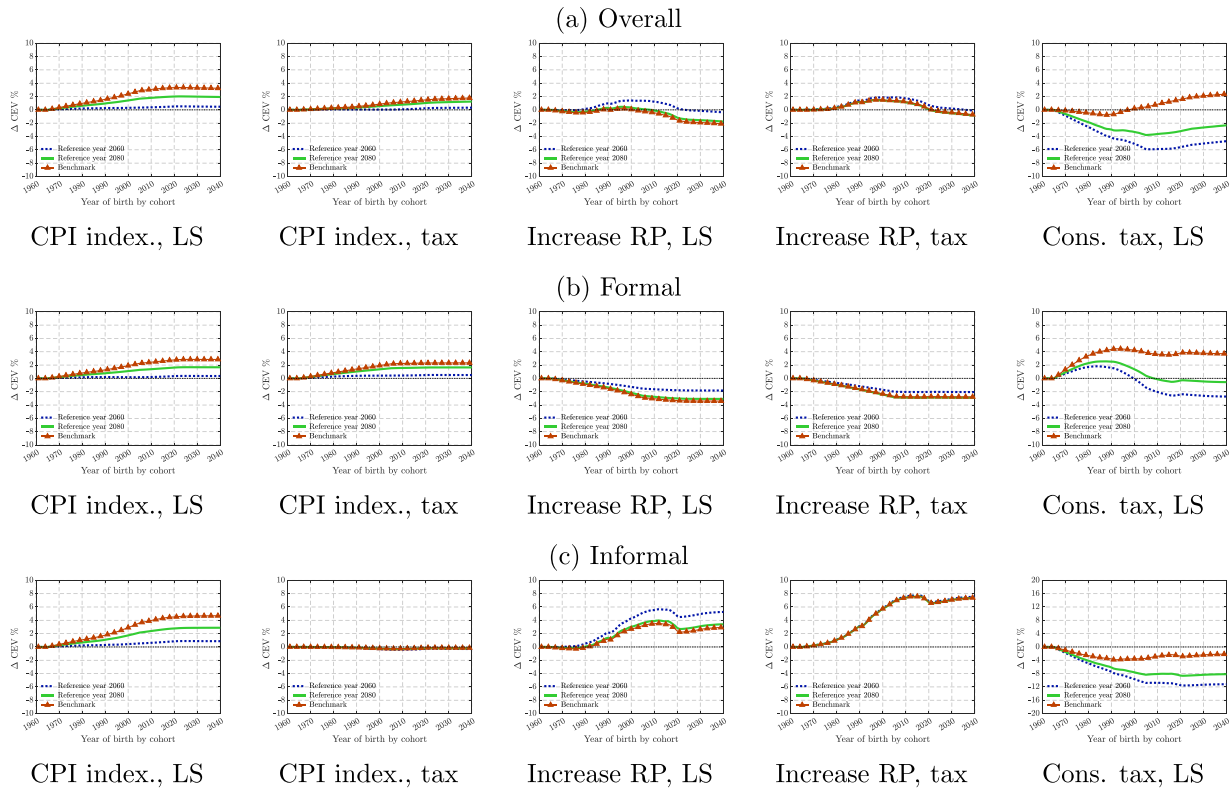


Fig. 13. Impact on welfare under alternative reference years of government budget balance.

Note: The red solid lines with triangular markers refer to results under baseline reference year of 2100, the blue dotted lines refer to results under the reference year 2060, and the green solid line refer to results under the reference year 2080.

Furthermore, agents reduce their labor supply due to the positive wealth effect of lump-sum transfers.

6. Sensitivity of results

In order to test the robustness of our findings for the baseline model, we conduct robustness along several dimensions by relaxing the assumptions made for the baseline in terms of model features or parameter values. These exercises, strongly motivated by policy concerns, illustrate the role of each of the model elements and provide an error bound of our benchmark estimates.

6.1. Sensitivity to reference year of balancing government budget

In the benchmark, we set the reference year for the government to balance the budget to 2100, before which the government can issue bonds at the market rate to smooth fluctuations in government revenue across different years. In this section, we consider two alternatives, 2060 and 2080 as reference years.

As shown in Fig. 9, the cost-saving effect of options 1 and 3 is more significant in the long run. Therefore, using a reference year in a shorter horizon will lead to lower lump-sum transfers or lower contribution tax cuts to maintain the government budget balanced. For example, under option 1, the lump-sum transfer is annually 3098 CNY at the wage level of year 2023 using reference year 2100, which reduces to 2005 CNY if using 2080 and further reduces to 775 CNY if using 2060. If alternatively implemented with adjustment of contribution tax rates, the contribution rate reduces to 19.2% using reference year 2100, but only to 19.6% using reference year 2080 and to 20.3% using reference year 2060. For the same reason, the size of negative transfer or the increase in contribution rate also decreases under option 2 where the government needs to increase revenue to finance the expansion of resident pension.

Since using an alternative reference year only involves the size of the policy to rebalance the government budget, the change in welfare effects hinges on the direction in which the rebalancing policy is headed, as shown in Fig. 13. Compared to using 2100, the welfare gains are smaller for option 1 using alternative reference years. Similarly, the welfare gain for the informal workers and the welfare loss for formal workers are smaller using alternative reference years, which drives up the overall welfare gain to positive if reference year 2060 is used. As shown in Fig. 9, it is only until 2055 that the consumption tax scheme makes a difference in the government budget compared to the baseline policy. As a result, the effect of distortion on intertemporal allocation dominates if an earlier reference year is used, making the consumption tax least favorable among all options. As can be expected, the impacts on household behavior move in the direction under which a smaller scale of rebalancing policy is used and are qualitatively similar.

6.2. Sensitivity to fertility rate projection

In our benchmark, we take the demographic process projected by the UN under the assumption of the medium fertility scenario. However, as the fertility rate in China hits its historically low level in recent years, doubts are raised as to whether the fertility rate can bounce back to 1.35% in the long run, as projected by the medium fertility scenario which is represented as the blue solid line in Fig. 14(a).

A lower fertility ratio leads to an increase in the dependency ratio of the population and a rapid decline of the total population, exacerbating the sustainability of pension policy. To address this concern, we present results under the low fertility scenario. As shown in the dashed red line of Fig. 14(a), the fertility rate is projected to increase only to 0.8% in the long run. As Fig. 14(b) shows, the population in 2100 is projected to be 400 million, which is only around 2/3 under the medium scenario.

In Fig. 15, we plot the impact of policy reforms on the government budget in both the benchmark and the low fertility scenarios. Panel (a)

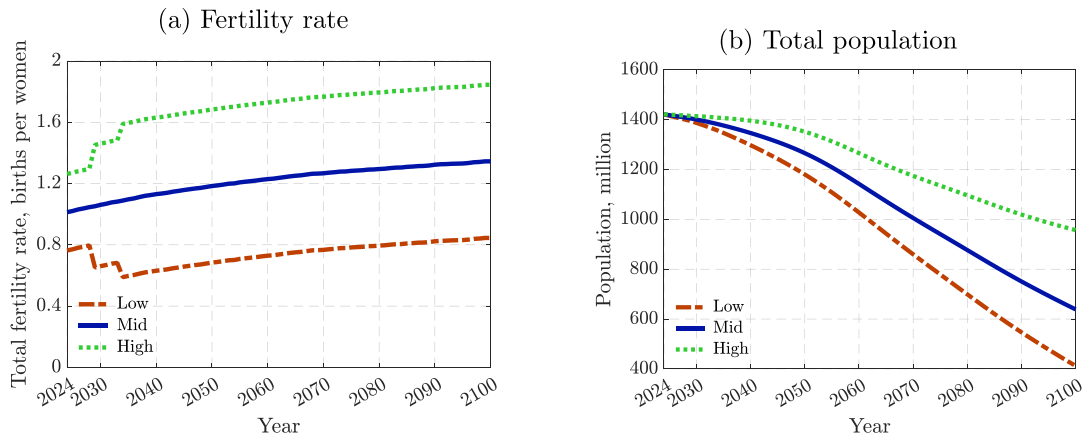


Fig. 14. WPP projection fertility variants.

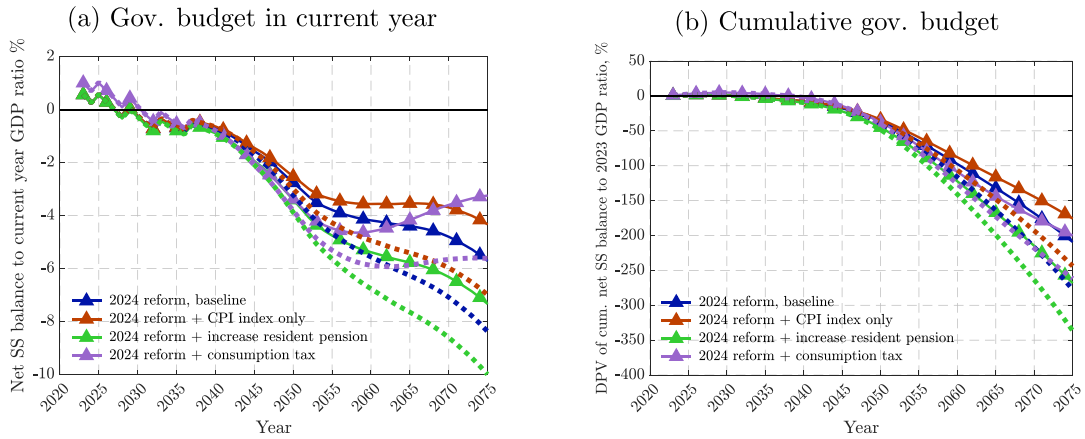


Fig. 15. Impact on government budget under alternative low fertility scenario.

Note: (i) The cumulative present value of public pension deficits is discounted using the market deposit rate. (ii) The solid lines with triangle markers refer to the results under baseline model, and the dotted lines are results under the alternative model assumptions.

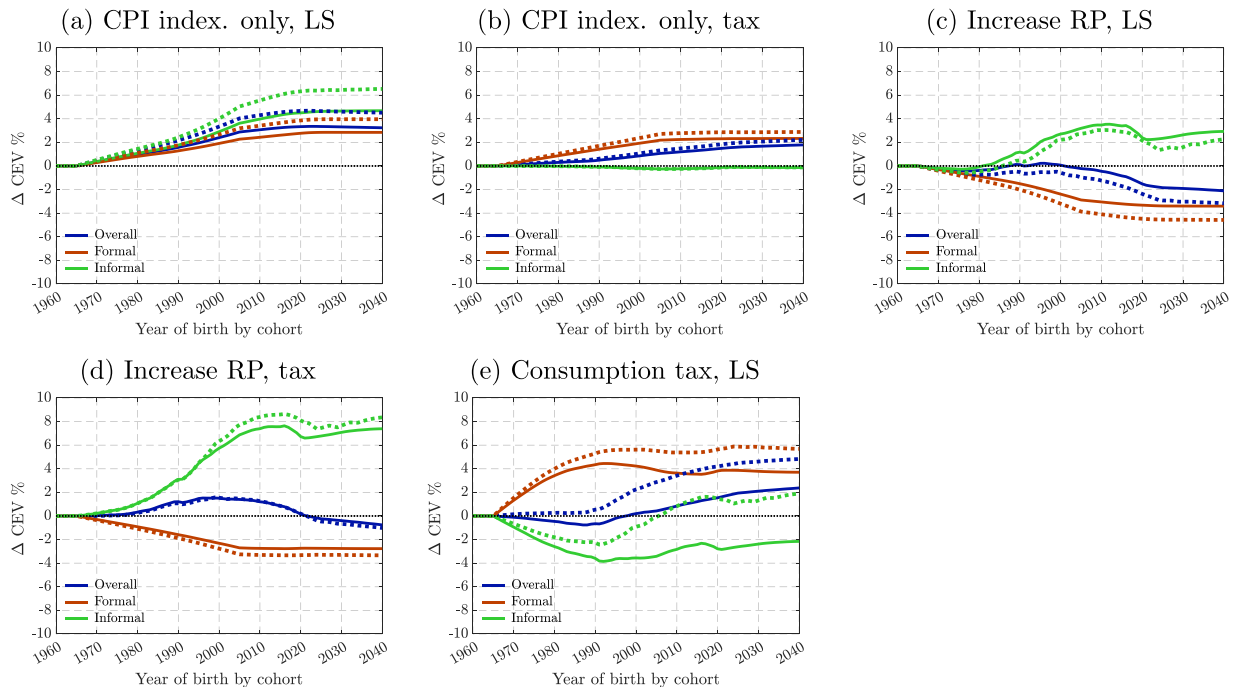


Fig. 16. Impact on welfare under low and medium fertility scenarios.

Note: The solid and dotted lines refer to the case of medium and low fertility variants, respectively.

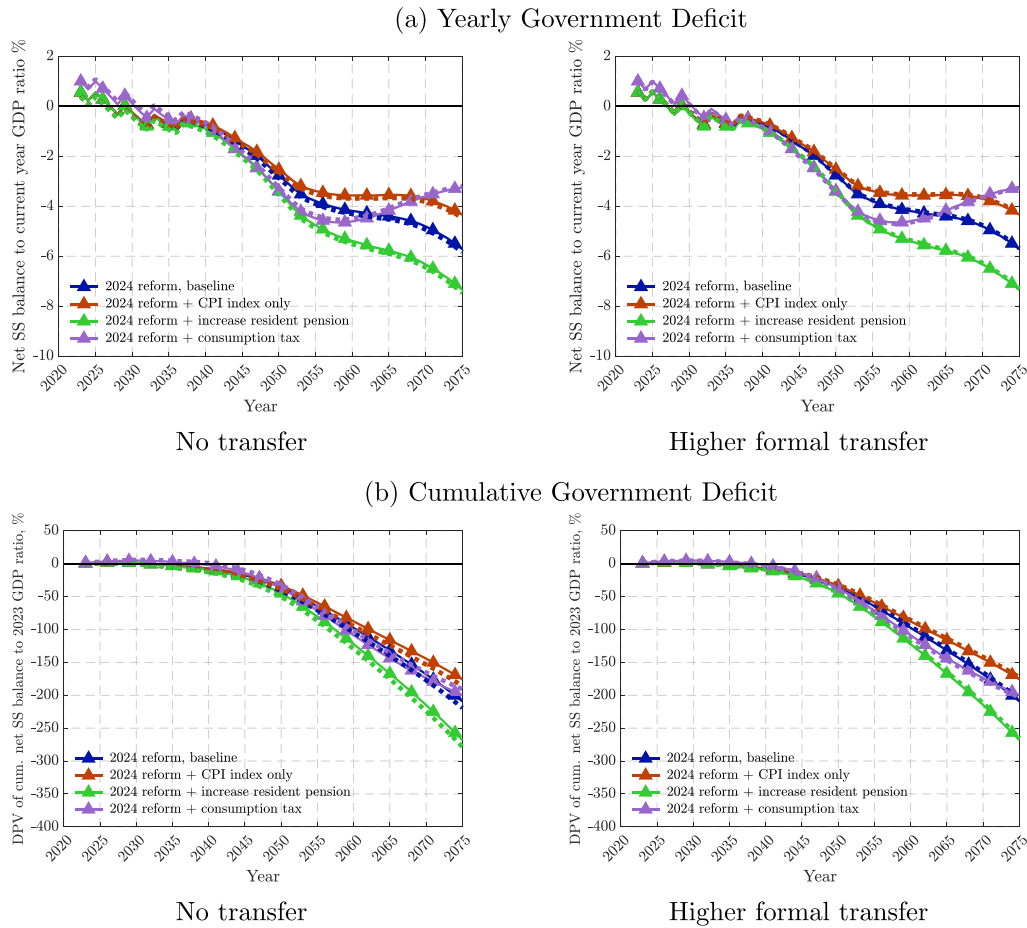


Fig. 17. Impact on government budget under alternative assumptions of intergenerational transfers.

Note: (i) The cumulative present value of public pension deficits is discounted using the market deposit rate. (ii) The solid lines with triangular markers refer to the results under baseline model, and the dotted lines are results under the alternative assumptions of intergenerational transfers.

presents the impact on the government deficit in each year, and panel (b) presents the impact on the cumulative deficit. As Fig. 15 shows, low fertility would exacerbate the government deficit to a large extent. For example, on an annual basis, the government deficit increases from 5.8% to 8% of GDP in year 2075 when we assume a lower fertility instead of the benchmark fertility scenario, as the comparison between the solid blue line with triangle marker and the dashed blue line shows. As a consequence, the cumulative deficit increases from 200% to 270% of GDP. In the low fertility scenario, the “revenue balancing” contribution rate increases to 49%. Although the magnitude is much larger under the low fertility scenario, the impact of policy reforms has the same direction between the two fertility scenarios. To be more specific, indexing benefits to CPI only and the consumption tax-based financing reduce government deficits, while increasing the benefits for the informal workers worsens the government budget.

Given the considerably worse fiscal trajectory of the pension system under the low fertility scenario, the magnitude of fiscal savings from the simulated reforms is larger under the low fertility scenario than the medium. However, the broad direction and pattern of policy reform impacts is similar between the two fertility scenarios.

We compare the welfare impacts of each reform option individually in medium and low fertility scenarios, both overall and for formal and informal sectors separately in Fig. 16. As in the baseline, removing the wage indexation of pension benefits and the consumption tax leads to the large welfare gains among all options, while increasing the resident pension leads to minor overall welfare changes, as it leads to changes in opposite directions for formal and informal workers. Our results illustrate that the welfare effects of the pension reforms

are qualitatively similar between the benchmark and the low fertility scenarios.

6.3. Sensitivity to intergenerational transfers

In our baseline, we assume that agents transfer part of their income to their parent generation when they are young and receive transfers from their children generation when they are old. To shed further light on the role of intergenerational monetary transfers, we consider two alternatives. First, we assume that there are no intergenerational transfers between generations. As informal workers rely more on intergenerational transfer when they get old, this is likely to have a major impact on their savings behavior and welfare. Second, instead of assuming that the transfer-to-income ratio is 9% for both formal and informal workers, we assume that formal workers transfer a higher fraction of income to their parents. To be more specific, we assume that formal workers transfer 12% of their income, while informal workers only transfer 6% of their income. This check aims to provide a better understanding of the relative importance of transfers for formal and informal workers in their savings and labor supply decisions when they are young.

In Fig. 17, we present the impact of reforms on the government budget. The left panel of panel (a) shows that removing intergenerational transfers leads to an almost negligible impact on the annual deficit. In panel (b), we further show that the impact on cumulative deficit is also very small. In the left panels of panel (a) and (b), we present results under the assumption of larger intergenerational transfer obligations

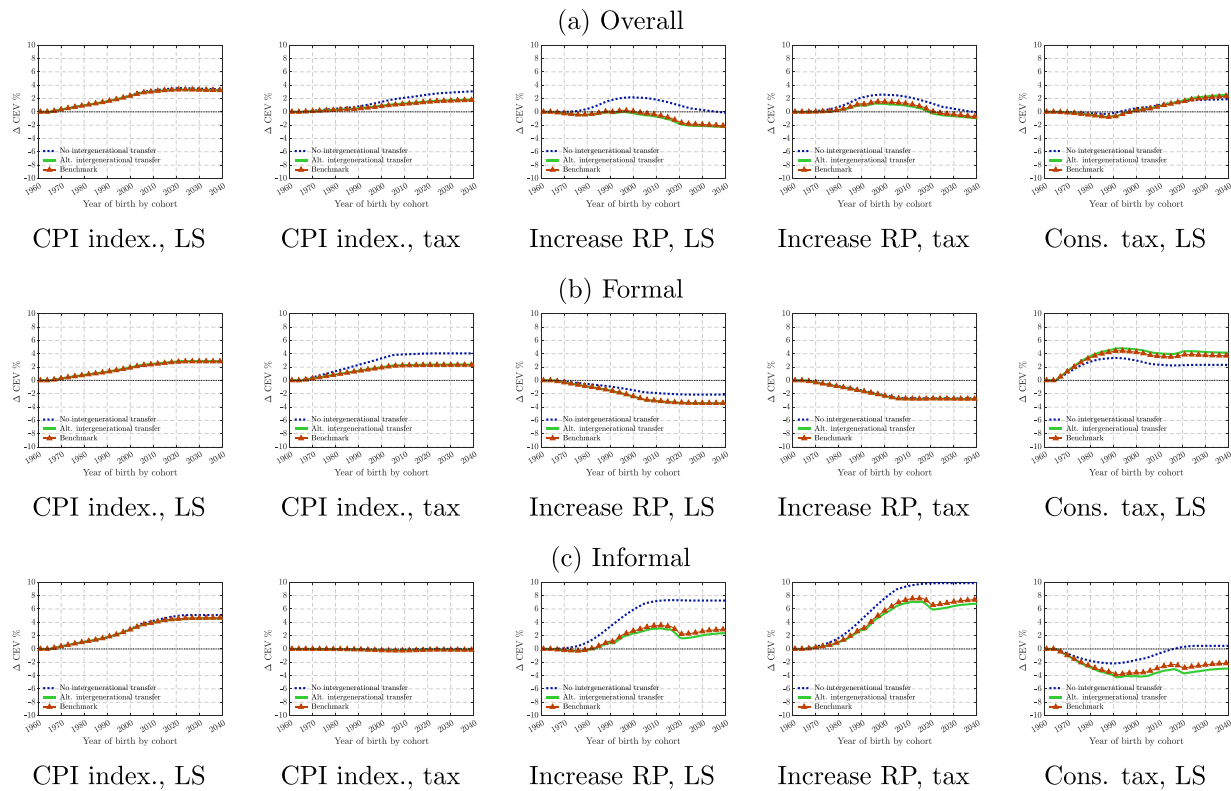


Fig. 18. Impact on welfare under alternative assumptions of intergenerational transfers.

Note: The red solid lines with triangular markers refer to results under baseline model, the blue dotted lines refer to results without intergenerational transfers, and the green solid line refer to results under the assumption of larger intergenerational transfers made by formal workers.

for formal workers. As both figures show, this assumption affects the projection of government deficit to a lesser extent.

Although the assumption of intergenerational transfers has little impact on the government budget, it plays an important role in affecting the evaluation of welfare effects. To be more specific, once the intergenerational transfers are shut down, agents, in particular the informal agents, are subject to a large loss of old-age incomes. Therefore, the welfare effects for informal workers will be amplified.

In panels (a), (b), and (c) of Fig. 18, we present the welfare impacts for the whole population, the formal population, and the informal population, respectively. Within each panel, we list five columns, each corresponding to a specific reform option and a specific way to maintain government budget balance. In all figures, we use the solid red line with triangle markers for results under the baseline assumption of transfers, the dotted blue line for results without transfers, and the solid green line for results with larger transfer obligations for formal workers.

As panel (a) shows, the overall welfare gain increases for all options when intergenerational transfers are shut down, in particular when the resident pension is increased, indicating the importance of incorporating intergenerational transfers into the framework. Furthermore, panel (b) shows that when there are no transfers, the policy experiments yield slightly larger government surplus for the scenario of CPI indexation and raising resident pension, leading to small welfare gains for formal workers. However, the welfare gain under the consumption tax is reduced. The most significant welfare changes occur in informal workers, as panel (c) shows. This is because when intergenerational transfers are shut down, the old informal workers rely more on the public pension. Therefore, the long-run welfare gain increases from 3% to 7.5% if the increase in resident pension is financed with a lump-sum tax, and from 7% to 10% if it is financed with an increase in the contribution rate for formal workers. Interestingly, due to the relative more important role of pension income for old-age support, the welfare

impact of consumption tax for informal workers also turns positive in the long run when transfers are not modeled, as opposed to long-run welfare loss in the baseline model.

6.4. Sensitivity to long-run wage growth and internal rates of return

In this subsection, we relax two assumptions that may directly affect the fiscal burden. First, we relax the assumption that wage growth converges to 2% in the long run and consider alternatively a higher level 2.5% and a lower level 1.5%. As pension benefits are only partially indexed to wage growth, a higher wage growth leads to lower pension budget deficits with a higher contribution relative to pension expenditure. Second, in our benchmark, we assume that the internal rate of return is the same for social-pooling account (SPA) and individual account (IA). Estimating the actual returns of the two accounts is empirically challenging, as the rates differ by region and are only available for a limited time window. In Section B of the Online Appendix, we present data on the returns and show that there is a trend of convergence in the returns to these accounts. To understand how the assumption of returns may affect our results, we alternatively consider two cases where the return to IA (R_I) is different from return to SPA (R_S). To be more specific, we calibrate separately for the return to IA and SPA so that the return to IA is 20% lower or higher than the SPA when the total return — or equivalently, the total pension incomes — for cohorts who are pension recipients in 2024 are controlled the same as in the baseline.

A lower long-run wage growth will decrease the contribution of the latter cohort, exacerbating the pension fund deficits under a PAYG system. As column 1 of panel (a) in Fig. 19 shows, under the new policy in 2024 without further reform implemented, when the long-run wage growth rate is 1.5%, the annual deficit is projected to be 8% of GDP, as opposed to around 6% in the baseline under the assumption of a 2% long-run growth. In panel (b) column 1, we show that the

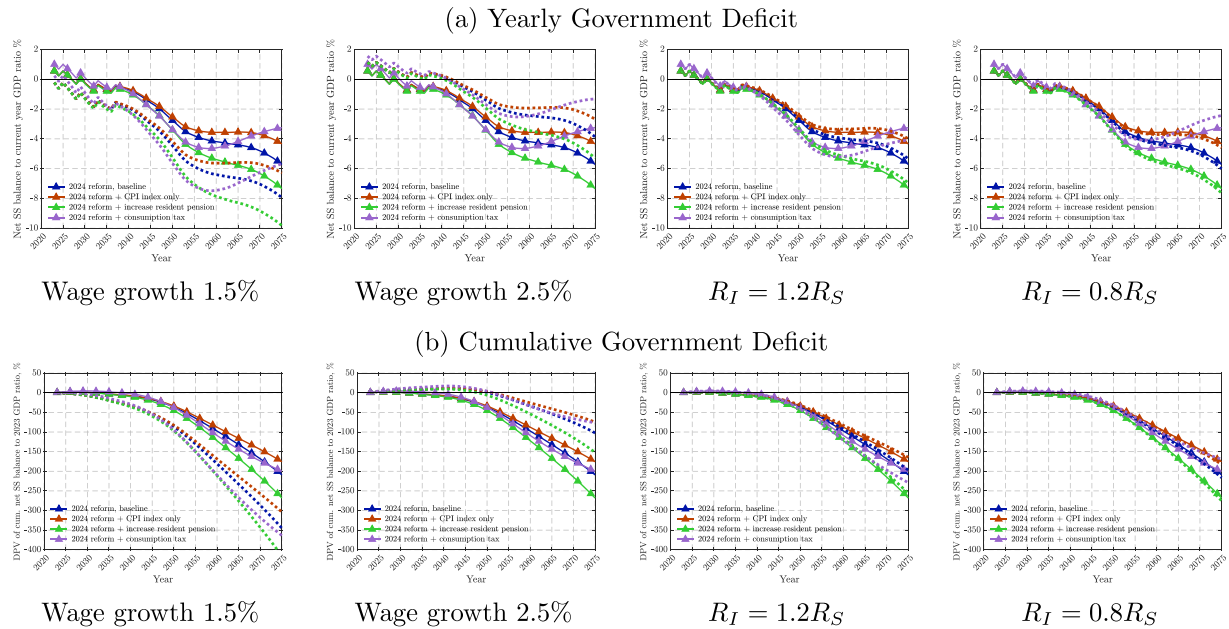


Fig. 19. Impact on government budget under alternative assumptions of long run wage growth and internal returns.

Note: (i) The cumulative present value of public pension deficits is discounted using the market deposit rate. (ii) The solid lines with triangular markers refer to the results under baseline model, and the dotted lines are results under the alternative model assumptions.

cumulative deficits will increase to 350% of GDP if the 1.5% wage growth is assumed. In addition, lower long-run wage growth will lead to slightly smaller pension fund savings with reform options 1 and 3 and larger deficits with reform 2. In contrast, when long-run wage growth is 2.5%, the public pension system becomes more sustainable. As panel (a) column 2 shows, the annual deficit will reduce to 4% by the year 2075. Panel (b) column 2 further shows that the public pension system will run a net cumulative surplus until 2050, and the cumulative deficit is only 100% of GDP by year 2075.

While the assumption of long-run wage growth has a large impact on the sustainability of public pension, the assumption of internal return of pension accounts appears to affect pension deficits to a lesser extent, as shown in columns 3 and 4 of Fig. 19. Yet, it still has important implications for the choice of further reform options. When the return of IA is 20% higher than the SPA, the public pension deficit is projected to be lower, since a larger share of pension funds have been invested in the social-pooling account. This happens in the cases where the 2024 policy is continued, or with further reform options 1 or 2. However, as the consumption-tax scheme changes the pension system to be complete individual account-based, the deficit under option 3 is assumed to be larger. The opposite happens when the return to the SPA is 20% higher than the IA, as Fig. 19 column 4 shows.

As in previous subsections, we present the welfare impacts for the whole population and the formal and informal population separately in Fig. 20. In all panels, the welfare change under the baseline model is depicted with a solid red line with triangle markers; welfare changes under the assumption of 1.5% and 2.5% long-run wage growth are depicted in solid and dotted blue lines, respectively; and welfare changes under the assumption of a higher return for IA and a lower return for IA are depicted in solid and dotted green lines, respectively. As shown in column 1 of Fig. 19, the assumption of lower wage growth will exacerbate the government deficit, therefore, suggesting greater cost savings for options 1 and greater deficit for option 3. As a consequence, the welfare gain is larger when option 1 is implemented, in particular if it is accompanied by a lump-sum transfer, as column 1 of Fig. 20 shows. For the same reason, the welfare loss is larger when the RP is increased, as columns 3 and 4 of Fig. 20 show. Column 5 shows that the welfare gain for consumption tax is smaller if a lower long-run wage

growth is assumed. Not surprisingly, the assumption of 2.5% long-run wage growth leads to welfare changes in opposite directions.

Interestingly, the assumption of internal returns for the SPA and the IA has a great impact on the welfare effects of the consumption tax but does not seem to have an impact on the welfare changes of options 1 and 2. To be more specific, under option 3 with consumption tax, as the government runs a larger deficit when the return to IA is assumed to be higher than SPA, the welfare gain is largely weakened, as the solid lines in column 5 show. In addition, this happens to both the formal and informal workers, leading to an overall welfare loss for cohorts born before 2040 and neutral welfare impacts for cohorts born after. In contrast, if the IA is 20% lower than the SPA, the consumption tax yields larger welfare gains, and all cohorts experience welfare gains. This exercise indicates the importance of pension account returns in evaluating the welfare effects of policy reforms.

6.5. Sensitivity to preference parameters

The final check involves the assumption of preference parameters in the model. In our baseline, we set the discount factor $\beta = 0.96$ and the risk aversion parameter $\sigma = 4$. As discussed in the previous section, the savings behavior of Chinese households may be explained by other factors that are not captured by our model. To illustrate how introducing these additional factors may affect the validity of our results, we consider alternative parameters for the preference. To be more specific, we consider a higher and lower level for the discount factor $\beta = 0.92$ and 1, and for risk aversion $\sigma = 2$ and 8.

In Fig. 21, we present the impact of policy reforms on the government budget. As in the previous sections, panel (a) presents the impact on the annual government budget and panel (b) presents the impact on the cumulative budget using the market rate. As can be seen in Fig. 21, the government deficit would reduce if we use a lower risk aversion ($\sigma = 2$) or a higher discount factor ($\beta = 1$) since both choices imply a weaker role for public insurance. The deficit will increase slightly if we use a higher risk aversion ($\sigma = 8$) or a lower discount factor ($\beta = 0.92$) although the effects are quite insignificant. In general, the preference parameters have a limited impact on the labor supply and thus a limit impact on the contribution and expenditure of the public pension funds.

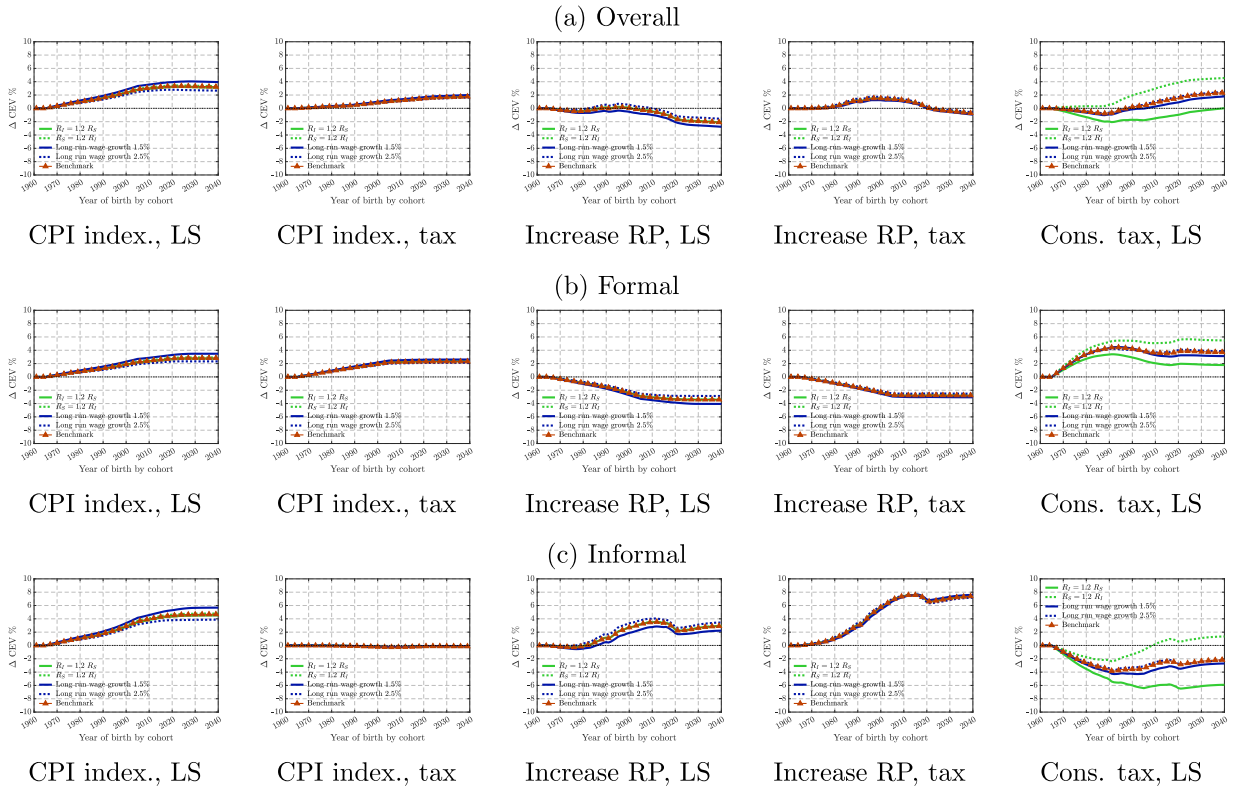


Fig. 20. Impact on welfare under alternative assumptions of long run wage growth and internal returns.

Note: The red solid lines with triangular markers refer to results under baseline model; the blue solid and blue dotted lines refer to results under the assumption of long run wage growth 1.5% and 2.5%, respectively; the green solid and green dotted lines refer to results under the assumption of the internal return of individual account is 20% higher or lower than the return of social-pooling account, respectively.

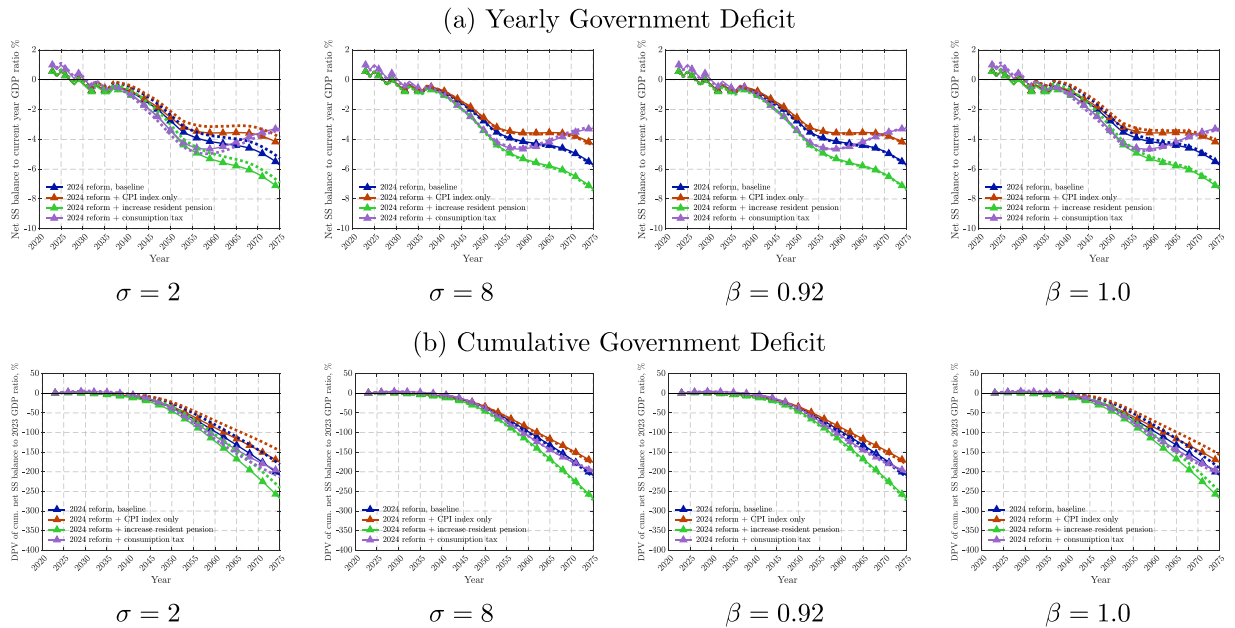


Fig. 21. Impact on Government budget under alternative assumptions of preference parameters.

Note: (i) The cumulative present value of public pension deficits is discounted using the market deposit rate. (ii) The solid lines with triangular markers refer to the results under baseline model, and the dotted lines are results under the alternative model assumptions.

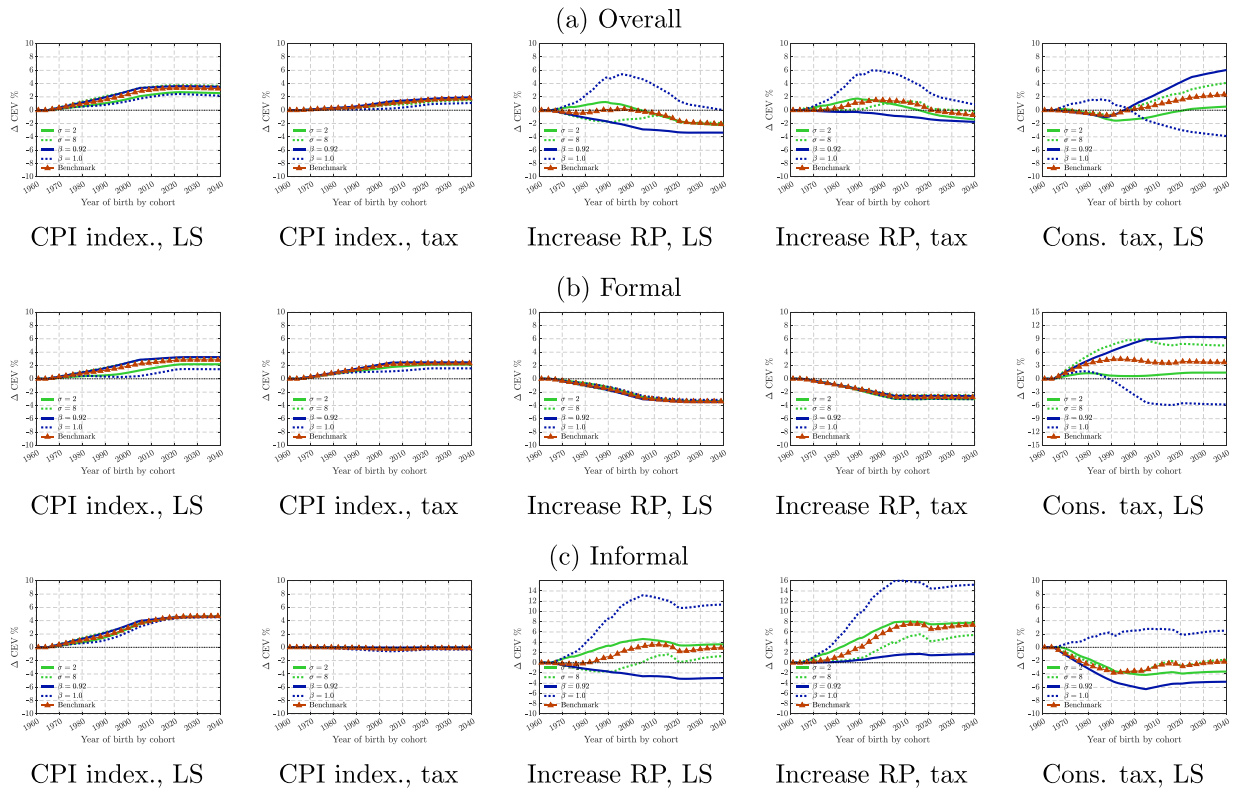


Fig. 22. Impact on welfare under alternative assumptions of preference parameters.

Note: The red solid lines with triangular markers refer to results under baseline model; the blue solid and blue dotted lines refer to results under the assumption of $\beta = 0.92$ and $\beta = 1$, respectively; the green solid and green dotted lines refer to results under the assumption of $\sigma = 2$ and $\sigma = 8$, respectively.

As a result, the choices of preference parameters do not seem to have a large impact on the government budget.

In Fig. 22, we next evaluate the welfare impacts of the reforms by the choice of preference parameters. As shown in columns 1 and 2, a lower discount factor or a higher risk aversion leads to slightly welfare gains, because both cases imply a stronger incentive for agents to receive transfer or contribution tax cuts as young as possible. Apparently, a lower risk aversion or a higher discount factor work in the opposite way. For option 2, the rebalancing of government budget comes in an opposite way, i.e., government needs to increase the contribution rate or use negative lump-sum transfers to finance the increase in resident pensions. As a result, a higher risk aversion or a lower discount factor will reduce welfare gains, in particular, for the informal sector. These effects are also magnified with slightly larger government deficits under a high risk aversion and a low discount factor, as shown in Fig. 21. The consumption tax works in a more complicated way. As discussed before, for formal workers, the consumption tax leads to distortion of intertemporal substitution and a reduction in pension contribution obligations, but the second effect dominates. These effects are amplified with a low discount factor and a higher risk aversion, leading to greater welfare gains, as column 5 of panel (b) shows. The opposite happens when a high discount factor or a low risk aversion is assumed. For the informal sector, a consumption tax means that they finance old-age benefits with contribution at younger ages. Therefore, welfare loss increases when they are more impatient with a lower discount factor or when they care less about uncertainty, i.e., when risk aversion is lower. The overall effects are dominated by the formal sector, as shown in column 5 of panel (a).

6.6. Discussion of other model assumptions

As a final remark, we discuss how we can address several elements that are not explicitly modeled in our framework. In Section 6.2,

we present the sensitivity of our results under the UN projection of the low fertility variant and show that both government deficits and welfare effects will be amplified. This, however, does not fully account for the endogenous nature of the fertility choice, in particular the quantity-quality trade-off of children (Becker and Lewis, 1973). Therefore, our model does not take into account the impact of dependent children on adult household members' labor supply and savings behaviors (Curtis et al., 2015; Choukhmane et al., 2023). A change in demographic structure can also lead to strategic behavioral responses such as competition in savings (Wei and Zhang, 2011), human capital investment (Choukhmane et al., 2023), and education (Kim et al., 2024). As these effects are interrelated, below we provide a brief discussion of the channels through which these may alter the results and how we can address these issues in our framework.

First, less generous Social Security benefits can encourage the supply of labor among older workers and the investment in human capital during the early stages of their careers, as shown by Iskhakov and Keane (2021) and Fan et al. (2024). In light of this mechanism, it is likely that option 1 where pension benefits are effectively reduced can stimulate human capital investment, which further improves the welfare effects. Through the lens of model, the impact on human capital, in particular, the extensive margin, can be addressed with an alternative projection of faster formalization of the population.

Second, behavioral responses in the human capital and asset market may propagate with an impact on factor prices of capital and labor through general equilibrium effects. This, however, depends on the quantitative strength of the incentive to accumulate physical or human capital. We provide sensitivity checks along two relevant margins, i.e., long-run wage growth in Section 6.4 and preference parameters in Section 6.5. We can extend the work once a better identification of the impact of reforms on aggregate factor supply is obtained.

7. Conclusion

In this paper, we construct a dynamic quantitative model with characteristics of Chinese economy to evaluate the long-run impacts of the public pension reform on government fiscal sustainability, household behaviors, and welfare. We also perform sensitivity checks to examine the role of various model elements in affecting the results, establishing the validity of our results.

We show that the 2024 reform can greatly reduce the fiscal financial burden and have a large impact on household savings and labor supply. We propose a few more reform options that aim to further address the sustainability and equity of the pension system. Gradually shifting the pension benefits to CPI indexation can save fiscal expenditure and lead to large welfare gains. Increasing the benefits of informal workers leads to overall welfare gains with better redistribution, but the gains diminish as the share of informal workers decreases in the total population.

The general applicability of results is established through various sensitivity checks, demonstrating the validity of results to rich environment and policy concerns. We show that if a low fertility rate becomes more likely, policies aimed at increasing social pension benefits should be approached with caution. Welfare improves when intergenerational transfers are reduced from their current levels. While higher wage growth helps contain the government budget, its welfare impact depends on policy design: it reduces welfare under CPI-indexed benefits but enhances it if social pensions are increased or if a consumption-based contribution mechanism is used. The policy setting for internal rate of return is also critical. Delaying retirement age without changing the social-pooling benefit formula effectively lowers the internal rate of return for social pooling relative to individual accounts. If individual accounts receive higher credited returns to preserve current benefit levels, this adjustment can reduce government deficits without significantly harming welfare. However, under a consumption-based contribution model, where all contributions fund individual accounts, a higher individual account credited return increases government deficits and reduces welfare.

It seems encouraging that the 2024 reform and the potential follow-up reform options can greatly improve the situation. However, there are still several dimensions that the model is overly simplified that need attention from both academic and policy perspectives.

First, it will take considerably longer for the impacts of long-run technological progress, in particular, automation, to emerge. It raises a concern for both researchers and policy makers to fully understand as fast automation can change factor prices, working opportunities, and working arrangements for workers, and thus the effectiveness of government policies. Furthermore, it will also have an impact on people's education and migration decisions, both of which have significant implications for the design of pension policy. Despite we have made original contributions in examining the effects through its impact on the demographic change (formality share) and factor prices (long-term wage growth), it still remains a question how to design a flexible pension policy to accommodate this change with more data and better understanding of the channels that technological progress affects household behavior.

Second, we simplified the compliance issue, in particular, from the firm side. However, in policy practice, there is a very strong sorting pattern between firm and worker, which implies that firms play an important role in increasing coverage to these informal workers. In this paper, we follow the convention of this field to introduce a representative firm. However, more efforts are needed to better understand how firms may respond to these reforms. We leave these avenues for future works.

Declaration of competing interest

The author acknowledge financial support from the Australian Research Council under grants DP210103319 and CE170100005 for the paper. The author declare that they have no relevant or material financial interests that relate to the research described in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econmod.2025.107336>.

Data availability

I have shared the link to my data and code at the attached file step.

Replication Package for Reforming China's Public Pension System: Fiscal Sustainability and the Challenge of Formality-Based Inequality (Original data) (Mendeley Data)

References

- Bairaliya, N., Miller, R., 2021. Social insurance, demographics, and rural-urban migration in China. *Reg. Sci. Urban Econ.* 91, 103615.
- Banerjee, A., Meng, X., Porzio, T., Qian, N., 2023. Aggregate fertility and household savings in China. Working Paper.
- Becker, G.S., Lewis, H.G., 1973. On the interaction between the quantity and quality of children. *J. Political Econ.* 81 (2, Part 2), S279–S288.
- Chamon, M.D., Prasad, E.S., 2010. Why are saving rates of urban households in China rising? *Am. Econ. J.: Macroecon.* 2 (1), 93–130.
- Chen, K., Wen, Y., 2017. The great housing boom of China. *Am. Econ. J.: Macroecon.* 9 (2), 73–114.
- Chomik, R., O'Keefe, P., Piggott, J., 2024. Pensions in aging Asia and the Pacific: Policy insights and priorities. ADB Economics Working Paper Series.
- Choukhmane, T., Coeurdacier, N., Jin, K., 2023. The one-child policy and household saving. *J. Eur. Econ. Assoc.* 21 (3), 987–1032.
- Coeurdacier, N., Guibaud, S., Jin, K., 2014. Fertility policies and social security reforms in China. *IMF Econ. Rev.* 62 (3), 371–408.
- Conesa, J.C., Kitao, S., Krueger, D., 2009. Taxing capital? Not a bad idea after all!. *Am. Econ. Rev.* 99 (1), 25–48.
- Conesa, J.C., Krueger, D., 1999. Social security reform with heterogeneous agents. *Rev. Econ. Dyn.* 2 (4), 757–795.
- Curtis, C.C., Lugauer, S., Mark, N.C., 2015. Demographic patterns and household saving in China. *Am. Econ. J.: Macroecon.* 7 (2), 58–94.
- Deng, Y., Fang, H., Hanewald, K., Wu, S., 2023. Delay the pension age or adjust the pension benefit? Implications for labor supply and individual welfare in China. *J. Econ. Behav. Organ.* 212, 1192–1215.
- Díaz-Giménez, J., Díaz-Saavedra, J., 2009. Delaying retirement in Spain. *Rev. Econ. Dyn.* 12 (1), 147–167.
- Esteban-Pretel, J., Kitao, S., 2021. Labor market policies in a dual economy. *Labour Econ.* 68, 101956.
- Fan, X., Seshadri, A., Taber, C., 2024. Estimation of a life-cycle model with human capital, labor supply, and retirement. *J. Political Econ.* 132 (1), 48–95.
- Fang, H., Feng, J., 2018. The Chinese pension system. NBER Working Paper No. w25088, National Bureau of Economic Research.
- Feng, J., He, L., Sato, H., 2011. Public pension and household saving: Evidence from urban China. *J. Comp. Econ.* 39 (4), 470–485.
- Gao, H., 2025. Social security and female labor supply in China. *J. Econ. Behav. Organ.* 238, 107203.
- Gao, H., McKiernan, K., 2023. Labor market sorting and public pensions in developing economies. Working Paper.
- He, H., Ning, L., Zhu, D., 2019. The impact of rapid aging and pension reform on savings and the labor supply: The case of China. IMF Working Paper, Vol. 2019, International Monetary Fund.
- Hernández, A., López, J., Galindo, F., Salas, F., 2017. Miles for retirement. Working Paper.
- Huang, W., Zhang, C., 2021. The power of social pensions: Evidence from China's new rural pension scheme. *Am. Econ. J.: Appl. Econ.* 13 (2), 179–205.
- Imrohoroglu, S., Kitao, S., 2009. Labor supply elasticity and social security reform. *J. Public Econ.* 93 (7–8), 867–878.
- Imrohoroglu, S., Kitao, S., 2012. Social security reforms: Benefit claiming, labor force participation, and long-run sustainability. *Am. Econ. J.: Macroecon.* 4 (3), 96–127.
- Iskhakov, F., Keane, M., 2021. Effects of taxes and safety net pensions on life-cycle labor supply, savings and human capital: The case of Australia. *J. Econometrics* 223 (2), 401–432.
- Jung, J., Tran, C., 2012. The extension of social security coverage in developing countries. *J. Dev. Econ.* 99 (2), 439–458.
- Kim, S., Tertilt, M., Yum, M., 2024. Status externalities in education and low birth rates in Korea. *Am. Econ. Rev.* 114 (6), 1576–1611.

- Kitao, S., 2014. Sustainable social security: Four options. *Rev. Econ. Dyn.* 17 (4), 756–779.
- Kitao, S., 2015. Fiscal cost of demographic transition in Japan. *J. Econom. Dynam. Control* 54, 37–58.
- Li, C., Lin, S., 2019. China's explicit social security debt: How large? *China Econ. Rev.* 53, 128–139.
- Lu, B., He, W., Piggott, J., 2014. Should China introduce a social pension? *J. Econ. Ageing* 4, 76–87.
- Meng, X., 2017. The labor contract law, macro conditions, self-selection, and labor market outcomes for migrants in China. *Asian Econ. Policy Rev.* 12 (1), 45–65.
- O'Keefe, P., Gao, H., Lu, B., Piggott, J., 2025. The dynamic evolution of China's pension system and future prospects and policy issues. Working Paper.
- Ruppert, K., Schön, M., Stähler, N., 2024. Consumption taxation to finance pension payments. *Econ. Model.* 130, 106570.
- Salditt, F., Whiteford, P., Adema, W., 2007. Pension reform in China: Progress and prospects. OECD Social, Employment and Migration Working Papers No. 53, OECD.
- Sin, Y., 2005. Pension Liabilities and reform options for old age insurance. World Bank Working Paper 2005-01, 1, pp. 1–71.
- Song, Z., Storesletten, K., Wang, Y., Zilibotti, F., 2015. Sharing high growth across generations: Pensions and demographic transition in China. *Am. Econ. J.: Macroecon.* 7 (2), 1–39.
- Wang, D., Feng, J., 2021. Chinese Pension Reform: Progress, Challenges and Prospects. World Bank Working Paper.
- Wang, X., Wen, Y., 2012. Housing prices and the high Chinese saving rate puzzle. *China Econ. Rev.* 23 (2), 265–283.
- Wei, S.-J., Zhang, X., 2011. The competitive saving motive: Evidence from rising sex ratios and savings rates in China. *J. Political Econ.* 119 (3), 511–564.
- Wheadon, D., Castex, G., Kudrna, G., Woodland, A., 2024. Non-linear means-tested pensions: Welfare and distributional analyses. *Econ. Model.* 138, 106759.